GEOTECHNICAL PROFESSIONALS INC.

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GEOTECHNICAL INVESTIGATION
PROPOSED RETAIL/RESIDENTIAL DEVELOPMENT
BELLA TERRA RESIDENTIAL
7777 EDINGER AVENUE
HUNTINGTON BEACH, CALIFORNIA

DRAFT

Prepared for:
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Project No. 3015.I

December 16, 2020

GPI GEOTECHNICAL PROFESSIONALS INC.

December 16, 2020

Bella Terra Associates LLC
6801 Hollywood Boulevard, Suite 170
Los Angeles, California 90028

Attention: Mr. Kristofer Golder
Senior Development Manager

Subject: Draft Report of Geotechnical Investigation
Proposed Retail/Residential Development
Bella Terra Residential
7777 Edinger Avenue
Huntington Beach, California
GPI Project No. 3015.I

Dear Mr. Golder:

Transmitted herewith is an electronic copy of our draft preliminary geotechnical investigation report for the subject project. The report presents our evaluation of the foundation conditions at the site and recommendations for design and construction.

We appreciate the opportunity of offering our services on this project and look forward to seeing the project through its successful completion. Feel free to call us if you have questions regarding our report or need further assistance.

Very truly yours,
Geotechnical Professionals Inc.

Donald A. Cords, G.E.
Principal

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1.0 INTRODUCTION

1.1 GENERAL

This draft report presents the results of the geotechnical investigation performed by Geotechnical Professionals Inc. (GPI) for the proposed Bella Terra Residential development in Huntington Beach, California. The site location is shown on the Site Location Map, Figure 1.

1.2 PROJECT DESCRIPTION

The proposed project will consist of a mixed-use retail and residential development. The proposed development will be located at the southwestern portion of Bella Terra retail center in a part of the area currently occupied by the Burlington Coat Factory building, adjacent single-story retail buildings, and a fountain area. The development will directly adjacent to the existing Bella Terra parking structure, existing retail, loading dock area, and drives. The project site is approximately 3.35 acres.

Based on a conceptual site plan provided to GPI, the proposed mixed-use retail and residential development will consist of a single 7-story building with a concrete podium. The ground floor podium will consist of a retail buildings, apartment lobby, and parking. The 2nd floor podium will consist of the parking only. The third floor will consist of a stepped parking podium, wood-frame apartments, an amenities room, and outdoor pool deck. The upper four floors will consist of wood-framed apartment buildings. A total of 300 apartment units are planned for the building. A basement is not planned for the development. At the ground floor, a 200 foot long driveway extends into the parking podium from the south. The drive continues through the new ground level parking and into the existing Bella Terra parking structure abutting the north side of the proposed development. The proposed site configuration is shown on the Site Plan, Figure 2.

We understand that the structure will consist of concrete podium for the first two levels. The third level will consist of a split concrete podium for the parking area and wood-frame construction for the apartment portion. We anticipate that the upper four stories will be wood-frame construction for the apartments. Based on the other multi-story buildings at Bella Terra, we anticipate that the ground floor will be constructed as a structural floor slab. Cross-sections of the proposed building are shown on Figure 3.

We anticipate that the finished floor of the building will be within 1 to 2 feet of the existing site grades.

Based on preliminary information provided by Mr. Zen Hoda of Englekirk, the Project Structural Engineer, the maximum column loads for the podium type building will be range from 630 to 860 kips (dead load and live load).

Since structural load or grade changes can significantly impact the performance of the proposed development, we should perform additional evaluations if the final grades and/or loads vary significantly from those discussed herein.

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1.3 PREVIOUS SITE INVESTIGATION

The site has been previously investigated by GPI. We performed several geotechnical investigations for the Bella Terra project. The results of our previous geotechnical investigations performed from 1999 to 2002, which include data near this project, are discussed in References 1 through 4. We performed a geotechnical investigation for the proposed Costco Warehouse (Reference 5) located directly northwest of this project and a geotechnical investigation for the retail/residential development located directly west of this project (Reference 6).

We have included the geotechnical data (including cone penetration tests, borings, and laboratory testing) from our previous investigations from explorations located near the project limits for the proposed Bella Terra Residential addressed in this report.

Prior to our investigations, Law Crandall (Reference 7) performed a geotechnical investigation in 1997 for the reconstruction of the retail center. Their investigation included 19 borings, 2 seismic cone penetration tests, and laboratory testing.

1.4 PURPOSE OF INVESTIGATION

The primary purpose of this investigation and report is to provide an evaluation of the existing geotechnical conditions at the site as they relate to the design and construction of the proposed development.

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2.0 SCOPE OF WORK

Our scope of work for this investigation consisted of review and use of existing geotechnical data, field exploration, laboratory testing, engineering analysis, and the preparation of this report.

We reviewed existing geotechnical investigations prepared by GPI (References 1 through 4). Information from these references has been incorporated herein, as appropriate. Data used in our report for the adjoining retail/residential development (Reference 6) has also been used in this report.

A total of ten Cone Penetration Tests (CPT's) and four exploratory borings have been performed within or just adjacent to the project site. A total of six CPT's and four borings were used for our previous investigations at Bella Terra performed from 1999 to 2010 (References 1 through 6). The locations of the explorations are shown on the Site Plan, Figure 2.

We have modified the numerical designation of the CPT's and borings from the previous reports for the Bella Terra development. Notes on the logs provide the original designation of the CPT's and borings as shown in our 2002 and 2010 geotechnical investigations (References 1 and 6).

The CPT's were advanced to depths ranging from 50 to 80 feet below existing site grades. Detailed logs of the CPT's and a summary of the equipment used are presented in Appendix A. The borings were drilled using rotary wash equipment to depths of 31 to 51½ feet below existing site grades. Details of the drilling and Logs of Borings are presented in Appendix B.

Shear wave velocity measurements were performed to a refusal depth of 80 feet below existing grade using a seismic cone penetration test. Details of the shear wave velocity measurements are presented in Appendix A.

Laboratory soil tests were performed on selected representative samples as an aid in soil classification and to evaluate the engineering properties of the soils. The geotechnical laboratory testing program included determinations of moisture content and dry density, grain size, compressibility (consolidation), shear strength (direct shear), expansion potential (expansion index), compaction (maximum density/optimum moisture), R-value, and corrosivity. Laboratory testing procedures and results are summarized in Appendix C.

Soil corrosivity testing was performed by M.J. Schiff & Associates under subcontract to GPI. R-value testing was performed by Geologic Associates under subcontract to GPI. Their test results are presented in Appendix C.

A site-specific response spectra for the site was generated by GPI. Details of our analysis are presented in Appendix D.

Engineering evaluations were performed to provide earthwork criteria, foundation, wall-below-grade, slab design parameters, and preliminary pavement sections. The results of our evaluations are presented in the remainder of this report.

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3.0 SITE CONDITIONS

3.1 SURFACE CONDITIONS

The site is bounded on the north by a 6-story parking structure, on the east by the loading dock area and 2-story retail building (Ulta Beauty), on the south by drives and parking, and on the west by drives, the Costco gas station, a 4-story retail/residential building, and an outdoor park and pavilion area. At the time of our field investigation, the site included the Burlington Coat Factory building constructed in the 1960's, single-story retail buildings constructed in 2004, and a fountain constructed with the retail buildings.

The site is relatively flat, with localized variations in surface grade. Existing ground surface elevations ranged from about +25 feet on the north side of the project to approximately +23 feet along drive south of the proposed project based on topography in utility plans for the original center remodel.

3.2 SUBSURFACE SOILS

Our field investigation disclosed a subsurface profile consisting of fill soils overlying native materials.

Fills were encountered to depths ranging between approximately 3 to 6 feet below existing grades in our borings. The fills consisted predominantly of silty sands, silty clays, and their mixtures. The consistency of these materials was loose to dense (sands) and soft (clays). Due to the method of drilling, determination of the depth of fill was difficult and the estimates on the boring logs should be considered very approximate. During the grading for the retail buildings in 2004, the fills were not removed in their entirety because the buildings were supported on concrete piles with structural floor slabs.

The natural soils consist of interbedded layers of organic silts and clays, clays, peat, silty sands, and sands. The peat deposits generally occur in layers up to 4 feet thick within the upper 20 feet of the soil profile, and occasionally occur in thinner layers interbedded within the organic silts and clays to depths of about 45 feet. The peat deposits are highly compressible. The organic silts/clays and peat are very soft to stiff, becoming stiffer with depth. The upper silty sand and sand layers are generally loose to medium dense and become dense to very dense at depths of 58 to 62 feet across the site.

The soils within the upper 30 to 40 feet are weak and compressible. The underlying soils become more dense and stiff with depth and exhibit moderate to high strength and moderate to low compressibility characteristics.

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3.3 GROUNDWATER AND CAVING

Groundwater was encountered at depths of approximately 6 to 10½ feet below existing grades in the borings surround the project site during our investigation performed in 2000. In the borings for the adjacent Costco and retail/residential development performed in 2008 encountered water at depths of 9 to 10 feet (References 5 and 6). In our recent CPT's, the holes caved at depths of approximately 7 to 10 feet, which is usually an indicator of groundwater depth. Due to the method of drilling, the potential for caving was very difficult to determine. The sandy soils are expected to cave severely below the groundwater.

The depths measured to groundwater are near the historical shallow groundwater levels. The historical high groundwater has been determined to be between 5 and 10 feet depth contours at the site by the State of California (Reference 8).

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4.0 CONCLUSIONS AND RECOMMENDATIONS

4.1 GENERAL

Based on the results of our investigation, it is our opinion that from a geotechnical engineering viewpoint it is feasible to develop the site as proposed. To help mitigate long term differential settlements of the proposed concrete podium structures, pile foundations and structurally supported floor slabs will be required.

The most significant geotechnical issues that will affect the design and construction of the proposed structures are as follows:

- Silty sand and sand layers between depths of 10 to 50 feet exhibit a potential for liquefaction. Should liquefaction of these layers occur, the estimated magnitude of induced settlement would be on the order of 2 to 3-inches. Differential settlement across 40 feet could be on the order of 1 to 1½-inches.

- The upper clays and peats encountered at the site are highly compressible. The placement of structural loads will cause significant long-term settlement. Therefore, we recommend that the concrete podium building for the project be supported on pile foundations penetrating the soft upper deposits and end-bearing in the underlying dense sandy soils.

- Overexcavation of the undocumented fills and upper compressible soils will not be required below the building pad for the concrete podium building since the structure will be required to be supported on pile foundations with a structural floor slab.

- The on-site clayey soils are expansive and will shrink and swell with changes in moisture content. The clayey soils should not be placed in retaining wall backfill or within 2 feet of flatwork or other concrete slabs-on-grade.

- Proprietary foundation systems, such as auger-cast pressure grouted displacement piles (APGD) are considered to be feasible for the project. This foundation system is designed, built, and installed by specialty geotechnical contractors. APGD piles have been used for the buildings recently constructed at Bella Terra about 8 years ago. The most suitable alternative will be based upon the economies of each system, the anticipated column loads, soil conditions, and environmental limitations, such as noise and vibration.

- We reviewed placing the multi-story concrete podium buildings on mat foundation as an alternative to pile foundations. We would anticipate static settlement of a mat foundation to range from approximately 1½ to 3 inches. We anticipate an additional potential seismic settlement due to liquefaction on the order of 2 to 3 inches. We estimate that the maximum total combined settlement (static and seismic) would be on the order of approximately 6 inches, which is beyond the typical performance of a mat foundation.

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- Surcharging the building pad prior to construction of a mat foundation could reduce static settlements. However, the height of the surcharge, trucking to construct, and the perimeter settlement impacts are not practicable with the adjoining retail and residential uses and significant utility lines surrounding the perimeter of the site.

- Ground improvement methods such as Geopiers, stone columns, and deep soil mixing are not feasible for technical reasons and impact to the existing residences and retail uses near the project site.

- The on-site soils are severely corrosive to ferrous metal. Foundation concrete should conform to a severe level of soluble sulfate exposure for soil.

Our recommendations related to the geotechnical aspects of the development of the site are presented in the subsequent sections of this report.

4.2 SEISMIC CONSIDERATIONS

4.2.1 General

The site is located in a seismically active area typical of Southern California and is likely to be subjected to strong ground shaking due to earthquakes on nearby faults.

We assume the seismic design of the proposed development will be in accordance with the California Building Code (CBC) 2019 edition. For the 2019 CBC, a Site Class D may be used based on the results our shear wave measurements in the seismic CPTs (Table A-1, Appendix A) performed at the site. The remaining seismic code values can be obtained directly from the tables in the building code using the above values and appropriate SEAOC/OSHPD web site (Reference 9).

The Project Structural Engineer should determine the seismic design method.

4.2.2 Strong Ground Motion Potential

Based on published information presented in Reference 10, the most significant fault in the proximity of the site is the Newport-Inglewood Fault, which is located about 5 kilometers southwest of the site.

During the life of the project, the site will likely be subject to strong ground motions due to earthquakes on nearby faults. Based on the SEAOC/OSHPD website (Reference 9), we computed that the site could be subjected to a peak ground acceleration (PGA_M) of 0.67g for a modal magnitude 6.7 earthquake. This acceleration has been computed using the mapped Maximum Considered Geometric Mean peak ground acceleration from ASCE 7-16 (Reference 11) and a site coefficient (F_{PGA}) based on site class. The predominant earthquake magnitude was determined using a 2-percent probability of exceedance in a 50-year period, or an average return period of 2,475 years. The structural design will need to incorporate measures to mitigate the effects of strong ground motion.

Based on our site specific response spectra analysis, we calculated a peak ground acceleration of 0.73g. The acceleration was calculated using a site-specific probabilistic response spectrum generated for the MCE per the requirements of ASCE 7-16 (Reference 11). The final probabilistic response spectrum was based on the geometric mean of the spectral response values at 5% damping for the three attenuation relationships noted in Table 1 of Appendix D. Details of the site specific response spectra analysis including site-specific seismic code values are included in Appendix D.

4.2.3 Potential for Ground Rupture

There are no known faults crossing or projecting through the site. The site is not located in an Alquist-Priolo Special Studies zone. Therefore, ground rupture due to faulting is considered unlikely at this site.

4.2.4 Liquefaction

Soil liquefaction is a phenomenon in which saturated cohesionless soils undergo a temporary loss of strength during severe ground shaking and acquire a degree of mobility sufficient to permit ground deformation. In extreme cases, the soil particles can become suspended in groundwater, resulting in the soil deposit becoming mobile and fluid-like. Liquefaction is generally considered to occur primarily in loose to medium dense deposits of saturated soils. Thus, three conditions are required for liquefaction to occur: (1) a cohesionless soil of loose to medium density; (2) a saturated condition; and (3) rapid large strain, cyclic loading, normally provided by earthquake motions.

The site is located within an area mapped as having a potential for soil liquefaction (Reference 12). Inclusion of a site on the maps does not mean that a hazard actually exists at the site. It simply means that the characteristics of the site require investigation of the hazard.

The California Building Code (CBC), ASCE 7-16 (Reference 11), and Special Publication 117A (Reference 13) requires that the ground motion used for liquefaction evaluation be based on the Peak Ground Acceleration (PGA_M) adjusted for site class effects. Accordingly, we used a ground acceleration of 0.67g for a magnitude 6.7 earthquake for our analyses. The potential for liquefaction was evaluated using the methods presented by the 2008 Idriss and Boulanger methodology (Reference 14) and modifications provided in Special Publication 117A.

To evaluate the potential for liquefaction at the site, we considered recent and historic groundwater levels. Ground water levels were encountered at depths of 6 to 10½ feet in our most recent borings. Prior explorations at the adjoining site encountered groundwater at a depth of 7 to 10 feet. We used a groundwater depth of 5 feet, corresponding to historical high level for the State for our evaluation.

The majority of the soils encountered in our CPT's and borings are cohesive. However, based on our evaluations, thin sand, silty sand and sandy silt layers between depths of 10 to 50 feet exhibit a potential for liquefaction. Should liquefaction of these layers occur, the estimated magnitude of induced settlement would be from approximately 2 to 3 inches. Differential settlement across 40 feet could be on the order of 1-inch to 1½ inches.

4.3 MITIGATION OF SETTLEMENT

The site soil profile includes compressible and liquefiable soils in the upper 35 to 50 feet. The placement of new fill or structural loads will cause significant long term settlement of these upper materials. Surface loads due to buildings will cause settlement of the compressible layers within the influence of the foundation loads. A design earthquake may cause liquefaction and dynamic settlement of a portion of these soils.

The potential building settlement under both static and earthquake loads could be mitigated by pile foundations. The pile foundations could consist of conventional driven concrete piles, APGD piles, or other proprietary pile types. The pile foundations would mitigate the static and dynamic settlement by extending below the liquefiable soils into a dense sand bearing material. The pile designs will need to account for the possible down drag forces due to dynamic settlement above liquefiable soils and static settlements of compressible soils layers if grades are raised significantly. Piles will need to extend to approximate depths of greater than 60 feet below existing grades to derive their support by friction and end bearing. Recommendations for pile foundations are provided in Section 4.5.

We reviewed placing the multi-story concrete podium buildings on mat foundations as an alternative to pile foundations. Mats can typically tolerate much greater total and differential settlements than conventional shallow foundations. A mat reduces the net load on the underlying soils by placing the mat below the existing grade and distributing the relatively high column loads over a large area. We would anticipate static settlement of a mat foundation to range from approximately 1½ to 3 inches. We anticipate an additional potential seismic settlement due to liquefaction on the order of 2 to 3 inches. The maximum total combined settlement of up to 6 inches is beyond the typical performance of a mat foundation.

The settlements of the concrete podium buildings due to structural loads from a mat foundation could be lessened by surcharging the building pad prior to building construction. However, due to the typical loads associated with a mat foundation, a surcharge will require importing of at least 10 feet of soil, will result in significant truck traffic, will require at least 60 to 90 days before settlement completes, and will induce settlement outside the surcharged area. This option is not feasible with the adjoining retail and residential uses and significant utility lines surrounding the perimeter of the site. In addition, surcharging will not significantly mitigate seismic settlements described Section 4.2.4.

We also reviewed ground modification methods to control settlements for the multi-story concrete podium building planned at the site. We reviewed typical methods used in Southern California such as rammed aggregate piers (Geopiers), vibro-replacement (stone columns) and deep soil mixing (soil-cement columns). Geopiers consist of drilled holes that are filled with aggregate base that is mechanically compacted as it is placed. Geopiers typically reduces the extent of remedial grading and anticipated static settlements. Vibro-replacement utilizes a large vibrating probe (mandrel) to create a cavity which is filled with gravel or crushed stone, and compacted as the mandrel is removed. The result is a stone column with the stone pushed laterally into the soil. The stone columns provide lower compressibility and higher shear strength while improving drainage. Stone columns can reduce both anticipated static and seismic settlement. Deep soil mixing involves the creation of soil/cement mixed columns extending through the soft compressible soil

deposits. The resultant is similar to that of stone columns in that the method results in lower compressibility and increased shear strengths. Deep soil mixing can reduce both anticipated static and seismic settlement.

Geopiers will not be effective at this site due to limitations in the depth of installation and their inability to densify surrounding soils to mitigate dynamic settlement due to liquefaction. Stone columns are generally effective if the soils to be densified are sands containing less than 15 to 20 percent fines. Significant liquefiable soils layers as discussed in Section 4.2.4 consist of silty sands with fine contents nearer to 35 percent. Past evaluations for the site indicated vibro-replacement will not mitigate the potential dynamic settlement due to liquefaction. Deep soil mixing can be ineffective in soils with significant organics. Past evaluations for the site indicated that the amount of cement or multiple treatments required within highly organics layers would make this method cost-prohibitive. Stone columns and deep soil mixing are also construction methods that cause significant ground water intrusion upward to the ground surface as well as noise and vibration. These construction activities may with noise and vibration that would impact the nearby residents and retail businesses.

4.4 EARTHWORK

The earthwork anticipated at the project site will consist of clearing, overexcavation of existing fills and natural soils, and soils disturbed by demolition activities, subgrade preparation, and placement and compaction of fill.

4.4.1 Clearing

Prior to grading, the areas to be developed should be stripped of vegetation, pavements, foundations, and cleared of all debris. Buried obstructions, such as utilities and tree roots, should be removed.

The existing buildings at the site are likely supported on concrete piles or Raymond step-tapered piles. Based on our experience with past demolition at the retail center, the buildings constructed in the early 1960's, such as the Burlington Coat Factory, are likely the Raymond step-tapered piles. The step-tapered piles consist of a conical steel shell driven into the ground and filled with concrete. The 1-story retail buildings constructed in 2004 are supported on 12-inch square, precast, prestressed concrete piles. The 12-inch piles are embedded to depths of about 53 to 59 feet. In general, the piles are contained in pile caps of 3 piles spaced at about 3 feet on center. GPI prepared a report of the pile installation for the retail buildings (Reference 15), which shows the approximate location of the piles.

We recommend the piles be removed to a depth of at least 2 feet below the bottom of the proposed structural floor slab for concrete podium building. We recommend that the existing piles be left in place and that the location and depth of the piles be surveyed during demolition. The results of the survey be provided to the Structural Engineer designing the foundations of the new structure. Deep foundations for the new structure may need to be adjusted based upon existing pile locations. The resulting excavation should be backfilled as recommended in the "Subgrade Preparation" and "Placement and Compaction of Fill" sections of this report.

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Although none were encountered, any cesspools or septic systems exposed during construction should be removed in their entirety. The resulting excavation should be backfilled as recommended in the "Subgrade Preparation" and "Placement and Compaction of Fill" sections of this report. As an alternative, any cesspools can be backfilled with a lean sand-cement slurry. Deleterious materials generated during the clearing operations should be removed from the site. At the conclusion of the clearing operations, a representative of the GPI should observe and accept the site prior to any further grading.

4.4.2 Excavations

Excavations at the site will include removal of unsuitable soils, foundation excavations and trenching for utility lines.

Within the building pad, overexcavation of the existing undocumented fills and upper compressible soils will not be required since the structure will be supported on pile foundations and a structural floor slab. Within drive areas to receive fill, overexcavation of the existing undocumented fills and upper compressible soils will not be required except where necessary to provide a stable bottom, capable of supporting compaction equipment.

For minor at-grade structures, such as trash enclosures and small site walls, we recommend removals of at least 2 feet below footing bottoms.

The actual depths of removal should be determined in the field during grading by a representative of GPI.

The removals should extend laterally beyond the edge of footings a minimum distance equal to the depth of overexcavation/compaction below finish grade (i.e. a 1:1 projection below the edge of footings).

Some of the near surface soils encountered in our explorations are very moist to wet. The use of rubber-tired equipment will most likely cause “pumping” and further disturbance of the subgrade soils. We recommend the contractor evaluate the moisture conditions and develop a method to prevent pumping of the subgrade soils exposed at the base of removals, such as the use of steel wheeled or track equipment within 3 feet of the subgrade soils. If the contractor uses rubber-tired equipment near the base of the removals instead of steel track, deeper removals may be required because of the resulting disturbance. We recommend that the contractor be held responsible for the costs associated with using unsuitable equipment (i.e., additional removals).

Temporary construction excavations may be made vertically without shoring to a depth of 4 feet below adjacent grade. For deeper cuts, the slopes should be properly shored or sloped back to at least 1:1 or flatter. Severe caving should be anticipated in excavations attempted below the groundwater level. As such, dewatering, shoring, excavation, and backfill methods should be developed by the contractor for structures or utilities that are anticipated to extend below the groundwater. Slopes as flat as 2:1 (H:V) may be required below groundwater to enhance excavation stability. Surcharge loads should not be permitted within a horizontal distance equal to the height of cut from the top of the excavation or 5 feet from the top of the slopes, whichever is greater, unless the cut is properly shored. Excavations that extend below an imaginary plane, inclined at 45 degrees below the edge of any adjacent existing site facilities, should be properly shored to maintain support of

adjacent elements. All excavations and shoring systems should meet the minimum requirements given in the most current State of California Occupational Safety and Health Standards.

4.4.3 Subgrade Preparation

Prior to placing fills or construction of the proposed structures, the subgrade soils should be scarified to a depth of at least 6 inches, moisture-conditioned, and compacted to at least 90 percent of the maximum dry density in accordance with ASTM D 1557. Moisture contents of the near surface soils encountered are typically 4 to 8 percent over-optimum.

Subgrade processing at the base of removals should not be performed in areas where soils exhibit high moisture contents. The purpose of this recommendation is to reduce the potential for subgrade disturbance.

In areas to receive pavements, the top 12 inches below the pavement base should be scarified, moisture-conditioned, and compacted to a minimum of 95 percent (90 percent for clayey soils) of the maximum dry density in accordance with ASTM D 1557.

Stabilization of the subgrade in the drive areas may be required if wet clayey soils are exposed without sufficient time in the schedule to disc and dry in warm weather. If subgrade preparation is planned to be performed during the rainy season, the likelihood of stabilization and/or weather-related delays should be expected for the subgrade preparation of the pavement areas.

Stabilization of subgrade soils exposed at depths greater than 2 feet below existing grades are expected to be required to facilitate the placement and compaction of fills. Stabilization of the subgrade will facilitate the support of heavy equipment or placement and compaction of fill (including aggregate base in pavement sections). For stabilization, 12 to 18 inches of compacted aggregate base over a geogrid (Tensar BX1100 or equivalent) will usually provide an adequate working surface. We suggest obtaining unit prices for stabilization of subgrade that is too soft and wet to allow for placement and compaction of overlying fills. Unit prices should be obtained for placement of 12 to 18 inches of aggregate base (see Section 4.13 for base requirements) over geogrid.

The aggregate base may be considered as part of the required compacted fill beneath slabs or the foundations for the minor at-grade structures.

4.4.4 Material for Fill

The surficial on-site soils are, in general, suitable for use as compacted fill with the exception of the on-site clays or organic silts. These soils should not be used behind retaining walls or immediately beneath on-grade concrete slabs where select or non-expansive fill is specified or recommended.

Provided it is acceptable to the reviewing governmental agencies and owner, crushed, inert demolition debris derived from the undocumented fill, existing pavements and building, may be used in fills with the following processing requirements:

- If the inert debris is crushed to a well graded mixture with maximum particle size of 1½ inches, the crushed material may be used directly in the fill without further blending.

- Inert debris up to a maximum size of 3 inches may also be used in fills, provided it is thoroughly blended with enough soils to form a well-graded mixture (typically a 3:1 soil to debris ratio).

Material with a maximum particle size of 1½ inches could also be used for stabilization of soft and wet areas expected in the planned overexcavations.

Imported fill material should be predominately granular (containing no more than 40 percent fines - portion passing No. 200 sieve) and non-expansive (Expansion Index of 20 or less). The import should also exhibit a minimum R-value of 30, consistent with the existing near surface soils. GPI should be provided with a sample (at least 50 pounds) and notified of the location of soils proposed for import at least 72 hours in advance of importing. Each proposed import source should be sampled, tested and accepted for use prior to delivery of the soils to the site. Soils imported prior to acceptance by the Geotechnical Engineer may be rejected if not suitable.

Soils used for compacted fills should not contain particles greater than 6 inches in size.

4.4.5 Placement and Compaction of Fills

Fill soils should be placed in horizontal lifts, moisture-conditioned, and mechanically compacted to at least 90 percent of the maximum dry density in building and pavement areas, in accordance with ASTM D-1557. In pavement areas, the upper 12 inches should be compacted to 95 percent (90 percent for cohesive soils). The optimum lift thickness will depend on the compaction equipment used and can best be determined in the field. The following uncompacted lift thickness can be used as preliminary guidelines.

Plate Compactors	4-6 inches
Track Equipment, Small Vibratory or Static Rollers (5-ton±)	6-8 inches
Scrapers and Heavy Loaders	8-12 inches

The maximum lift thickness should not be greater than 12 inches.

The moisture content of the on-site materials should be between 0 to 3 percent over the optimum moisture content to readily achieve the required degree of compaction. The on-site soils are generally well above the optimum moisture content such that some drying during grading will be required. The contractors should allow for drying of these materials and stabilization of the subgrade in their bids.

During backfill of excavations, the fill should be properly benched into the construction slopes as it is placed in lifts.

4.4.6 Shrinkage and Subsidence

Shrinkage is the loss of soil volume caused by compaction of fills to a higher density than before grading. Subsidence is the settlement of in-place subgrade soils caused by loads

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generated by large earthmoving equipment. For earthwork volume estimating purposes, an average shrinkage value of about 10 percent and subsidence of 0.1 feet may be assumed for the surficial soils. These values are estimates only and exclude losses due to removal of vegetation or debris. Actual shrinkage and subsidence will depend on the types of earthmoving equipment used and should be determined during grading.

4.4.7 Trench/Wall Backfill

Utility trench and wall backfill consisting of the on-site material or imported sand should be mechanically compacted in lifts. The on-site clayey or organic soils should not be used in retaining wall backfill and may be difficult to compact in trenches. Some drying of the on-site soils should be anticipated prior to backfill. Lift thickness should not exceed those values given in the "Compacted Fill" section of this report. A representative of GPI should observe and test trench and wall backfills as they are placed.

In backfill areas where mechanical compaction of soil backfill is impractical due to space constraints, sand-cement slurry may be substituted for compacted backfill. The slurry should contain at least 1½ sack of cement per cubic yard and have a maximum slump of 5 inches. Within the building area, the slurry should contain two sacks of cement per cubic yard. When set, such a mix typically has the consistency of compacted soil.

4.4.8 Observation and Testing

A representative of GPI should observe all excavations, subgrade preparation, and fill placement activities. Sufficient in-place field density tests should be performed during fill placement and in-place compaction to evaluate the overall compaction of the soils. Soils that do not meet minimum compaction requirements should be reworked and tested prior to placement of any additional fill.

4.5 PILE FOUNDATIONS

4.5.1 General

The proposed concrete podium building may be supported on piles to help mitigate excessive total and differential settlement.

Either precast, prestressed, concrete drive piles or Auger Pressure Grouted Displacement (APGD) piles are considered the most feasible to support the building. However, due to the noise and vibration associated with driven piles, we understand that driven piles may not an acceptable alternative with the nearby residential and retail businesses.

Other proprietary pile alternatives may be considered for the project. The most suitable pile alternative will be based on the economies of each system, the anticipated column loads, soil conditions, and environmental limitations, such as noise and vibration.

We provide recommendations for APGD piles below, which were used for the Costco and retail/residential buildings most recently in Bella Terra directly west of the proposed project.

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4.5.2 APGD Piles

APGD piles derive their support from a combination of friction and end-bearing in the dense to very dense sands at depth.

APGD piles are installed by drilling holes with an auger bit attached to a displacement element. The special auger bit displaces soil as it is advanced, creating increased capacity and minimizing drill spoils. Upon completion of drilling, grout is pumped under pressure from the auger tip as it is withdrawn to help maintain sidewall integrity, reducing the potential impact of caving soils. APGD piles are more flexible than driven piles with respect to pile length and the potential need for pile cut-offs when refusal in very dense materials is encountered. Also, APGD piles create significantly less noise and vibration during installation as compared to driven piles.

The APGD piles are typically designed, built, and installed by specialty pile contractors. All aspects of the design, construction, and performance verification of such systems are the responsibility of the registered engineer designing the system. In addition, foundation design parameters, including allowable capacities and estimated settlements must be provided by the designer.

Axial Capacity

Based on our recent field explorations, we recommend using allowable pile capacities for APGD piles end bearing into dense to very dense sands at depths of greater than 60 feet below existing grade at the site.

The following estimates for axial and lateral design capacities are provided for information only to develop preliminary cost estimates for the structures:

Preliminary Estimate of
Pile Embedment and Allowable Capacity
16-inch Diameter APGD

Pile Tip Below Existing Grade (ft)	Allowable Axial Pile Capacity (kips)
62	145

The allowable pile capacities presented above include a factor of safety of 2 after reduction of ultimate capacities due to downdrag caused by liquefaction. The allowable capacities account for a downdrag force of 45 kips for 16-inch APGD piles. Group action may be neglected provided the piles are spaced at least 3 pile diameters apart.

To resist uplift loads, the piles will derive their resistance from friction between the subsurface soils and the pile surface. Preliminary uplift capacity of the APGD piles may be taken as ½ the downward capacities presented above.

Pile Settlement

Preliminary estimates of the total settlement of the more heavily loaded interior columns supported by APGD piles end bearing into the dense sands under a maximum total load of approximately 850 kips, is estimated to be less than ½-inch. Differential settlements between similarly loaded adjacent columns are expected to be less than ¼-inch.

Lateral Capacity

We analyzed the lateral load response of 16-inch diameter APGD piles using LPILE, a finite difference computer program. The soil is modeled as a series of non-linear lateral springs, with lateral response defined by segmentally-variable p-y curves. Pile deflection with load is assumed to follow a linear-elastic relationship. Therefore, the results of our analyses are valid up to the bending moment capacity of the pile. At higher loads, the deflections will be greater than those predicted by the computer model.

The lateral capacity of the piles will depend on the permissible deflection and on the degree of fixity at the top of the pile. The capacities have been determined for allowable lateral deflections of ¼- and ⅜-inch and free and fixed head conditions at the top of the pile. The lateral capacities for the above-mentioned pile diameter are presented in the following table.

Preliminary Estimate of Lateral Capacity

PILE SIZE/ TYPE	LATERAL CAPACITY (kips) for ¼-inch DEFLECTION	LATERAL CAPACITY (kips) for ⅜-inch DEFLECTION
16” Diameter– Free Head	9	11.5
16” Diameter– Fixed Head	19	24

In Figures 4 and 5, we present lateral deflection, bending moment, and shear diagrams for the piles and deflections presented above.

For the L-PILE analysis, we assumed the upper 5 feet of soil would be sand with a unit weight of 115 pcf and a p-y subgrade modulus (k) of 120 pci. Below a depth of 5 feet, we assumed groundwater (historical high) and soft clay with a buoyant unit weight of 47.5 pcf and a cohesion of 750 psf. We assumed the modulus of elasticity of the concrete piles to be 3.8×10^6 psi.

In addition, lateral resistance will be provided by passive resistance against grade beams and pile caps. Assuming that these elements are poured tight against the existing ground, an allowable passive soil resistance equal to an equivalent fluid pressure of 300 pounds per cubic foot may be used. We recommend neglecting the upper 1 foot of the grade beams or pile caps when determining the available passive resistance to allow for future soil settlement.

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Pile Load Tests

The capacities of the APGD piles will be determined by the design/build contractor's Engineer and confirmed based upon pile load testing. We recommend that at least three APGD piles be load tested (2 downward and 1 sustained upward).

The load testing prior to installation of production piles is used to establish installation criteria. Measurements of drilling torque and grout volume are recorded throughout the installation process. GPI should continuously observe the installation of APGD piles for the load tests at the site. Upon evaluation of the pile load test results, GPI will either confirm the capacity or recommend a revised capacity based upon our independent evaluation.

Pile Installation

APGD piles are to be designed to derive their support from friction and end bearing. If APGD piles are used to support the concrete podium buildings, each pile should be installed to a design tip elevation based on load testing and a predetermined minimum drilling torque and grout volume criteria.

It is standard of practice for an independent inspector to observe the construction of a proprietary foundation system to confirm that the elements have been installed in accordance with the plans and specifications. To ascertain that piles are properly installed to sufficient depth to develop the required supporting capacities, it is imperative that a representative of GPI continuously observe the installation of APGD piles or other any other type of piles at the site.

We recommend that GPI review the final foundation plans and specifications to ascertain that the recommendations presented herein have been properly incorporated into the contract documents.

4.6 SHALLOW FOUNDATIONS

4.6.1 General

Minor structures with relatively light structural loads may be supported on shallow footings, provided the anticipated settlements (including both static and dynamic) are tolerable and the subsurface soils are prepared in accordance with the recommendations given in this report.

4.6.2 Allowable Bearing Pressures

Based on the shear strength and elastic settlement characteristics of the recompacted on-site soils, a static allowable net bearing pressure of up to 2,000 pounds per square foot (psf) may be used for both continuous footings or isolated column footings. These bearing pressures are for dead-plus-live loads, and may be increased one-third for short-term, transient, wind and seismic loading. The actual bearing pressure used may be less, such that economics and structural loads will determine the minimum width for footings as discussed below. The maximum edge pressures induced by eccentric loading or overturning moments should not be allowed to exceed these recommended values.

4.6.3 Minimum Footing Width and Embedment

The following minimum footing widths and embedments are recommended for the corresponding allowable bearing pressure.

STATIC BEARING PRESSURE (psf)	MINIMUM FOOTING WIDTH (inches)	MINIMUM FOOTING* EMBEDMENT (inches)
2,000	18	18
1,500	15	15

* Refers to minimum depth below lowest adjacent grade.

A minimum footing width of 15 inches should be used even if the actual bearing pressure is less than 1,500 psf.

4.6.4 Estimated Structural Settlements

Maximum total static settlement of lightly loaded footings is expected to be less than 1 inch. Maximum differential settlements between similarly loaded adjacent footings are expected to be less than ½-inch. These settlements should be included with the anticipated areal settlement caused by seismic settlement from liquefaction when evaluating the total settlement of the lightly loaded structures.

The above estimates are based on the assumption that the recommended earthwork will be performed and that the footings will be sized in accordance with our recommendations.

4.6.5 Lateral Load Resistance

Soil resistance to lateral loads will be provided by a combination of frictional resistance between the bottom of footings and underlying soils and by passive soil pressures acting against the embedded sides of the footings. For frictional resistance, a coefficient of friction of 0.35 may be used for design. In addition, an allowable lateral bearing pressure equal to an equivalent fluid weight of 300 pounds per cubic foot may be used, provided the footings are poured tight against compacted fill soils. These values may be used in combination without reduction.

The design of light pole bases will likely be governed by lateral force considerations. For design by the simplified pole formula presented in Section 1807A.3 of the 2019 CBC, a unit passive pressure of 600 pounds per square foot may be used for the pole bases in lieu of presumptive lateral bearing values presented in Table 1806A.2. This value incorporates the allowable increases stated in the code for single poles that can tolerate a ½-inch of deflection under short-term loads.

4.6.6 Footing Excavation Observation

Prior to placement of concrete and steel, a representative of GPI should observe and approve all footing excavations.

4.7 FOUNDATION CONCRETE

Laboratory test results by M.J. Schiff (now HDR) on a sample obtained from our explorations indicate the on-site soils have soluble sulfate content of 340 and 12,300 mg/kg (0.034 and 1.23 percent by weight). Based upon this information, foundation concrete for slabs, footings and piles should conform to the requirements outlined in ACI 318, Section 4.3 for a severe level of soluble sulfate exposure for soil.

4.8 BUILDING FLOOR SLABS

The pile supported structural floor slab should be designed by the Structural Engineer to support the anticipated loads.

A vapor/moisture retarder should be placed under new slabs that are to be covered with moisture-sensitive floor coverings (wood, vinyl tile, etc.). Currently, common practice is to use a 15-mil polyolefin product such as Stego Wrap for this purpose. Whether to place the concrete slab directly on the vapor barrier or place a clean sand layer between the slab and vapor barrier is a decision for the Project Architect, as it is not a geotechnical issue. If covered by sand, the sand layer should be about 2 inches thick and contain less than 5 percent by weight passing the No. 200 sieve. Based on our explorations and laboratory testing, the soils at the site are not suitable for this purpose. The function of the sand layer is to protect the vapor retarder during construction and to aid in the uniform curing of the concrete. This layer should be nominally compacted using light equipment. The sand placed over the vapor retarder should only be slightly moist. If the sand gets wet (for example as a result of rainfall or excessive moistening) it must be allowed to dry prior to placing concrete. Care should be taken to avoid infiltration of water into the sand layer after placement of the concrete slab, such as at slab cut-outs and other exposures. A sand layer is not required beneath the vapor retarder, but we take no exception if one is provided.

It should be noted that the material used as a vapor retarder is only one of several factors affecting the prevention of moisture accumulation under floor coverings. Other factors include maintaining a low water to cement ratio for the concrete used for the floor slab, effective sealing of joints and edges (particularly at pipe penetrations), as well as excess moisture in the concrete. The manufacturer of the floor coverings should be consulted for establishing acceptable criteria for the condition of floor surface prior to placing moisture-sensitive floor coverings.

4.9 LATERAL EARTH PRESSURES

Based on information available to us at the time this report was prepared, no major retaining walls or basements were planned on the site. The following recommendations are provided for walls less than 8 feet in height. We recommend that imported, non-expansive, granular soils be used as wall backfill.

Active earth pressures can be used for designing walls that can yield at least 1-inch laterally in 10 feet of wall height under the imposed loads. For level backfill comprised of granular soils, the magnitude of active pressures are equivalent to the pressures imposed by a fluid weighing 35 pounds per cubic foot (pcf). This pressure may also be used for the design of temporary excavation support.

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At-rest pressures should be used for restrained walls that remain rigid enough to be essentially non-yielding. At-rest pressures imposed by a fluid weighing 55 pounds per cubic foot should be used for granular backfill.

As outlined in the California Building Code, site retaining walls 6 feet or taller should be designed to resist seismic lateral earth pressures. A lateral pressure equivalent to a fluid with a unit weight of 22 pounds per cubic foot may be used. This pressure should be combined with the active earth pressure presented above. If the retaining walls are designed using the at-rest pressure provided above, only the difference between the active plus seismic pressures and the at-rest pressure needs to be included as the seismic pressure.

Walls subject to surcharge loads should be designed for an additional uniform lateral pressure equal to one-third and one-half the anticipated surcharge pressure for unrestrained and restrained walls, respectively.

The wall backfill should be well-drained to relieve possible hydrostatic pressure or designed to withstand these pressures. A drain consisting of perforated pipe and gravel wrapped in filter fabric should be used. One cubic foot of rock should be used for each lineal foot of pipe. The fabric (non-woven filter fabric, Mirafi 140N or equivalent) should be lapped at the top.

Wall footings should be designed as discussed in the "Shallow Foundations" section.

4.10 CORROSIVITY

Resistivity testing of representative samples of the on-site surficial soils by M.J. Schiff (now HDR) indicate that the soils are severely corrosive to ferrous metals (resistivity measurements of 970 ohm-cm). GPI does not practice corrosion engineering. Should the use of buried metal pipe be proposed, a corrosion engineer, such as HDR (formerly M.J. Schiff & Associates), should be consulted.

4.11 DRAINAGE

Positive surface gradients should be provided adjacent to all structures so as to direct surface water run-off and roof drainage away from foundations and slabs toward suitable discharge facilities. The introduction of water into the existing fill soils can result in subsidence. Long-term ponding of surface water should not be allowed on pavements or adjacent to buildings.

4.12 EXTERIOR CONCRETE AND MASONRY FLATWORK

Exterior concrete and masonry flatwork should be supported on non-expansive, compacted fill. The use of the clayey soils within 2 feet of the slab subgrade should not be permitted unless differential heave is tolerable. This includes exterior sidewalks, stamped concrete, non-traffic pavement, pavers, etc. Prior to placement of concrete, the subgrade should be prepared as recommended in the "Subgrade Preparation" section of this report.

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4.13 STORM WATER INFILTRATION

Current regulations require that storm water be infiltrated into the site soils of new developments when possible. The soil types present at the site control the ability of water to infiltrate into the subgrade.

The subsurface soils encountered above the measured groundwater consist of predominantly interbedded fine grained silts and clays and silty sands. The fine grained soils have very poor infiltration characteristics. The historic high ground water at this site is reported by the State at between 5 and 10 feet below existing grades. GPI has observed groundwater as shallow as 6 feet in borings surrounding the project site. The County's Technical Design Document (Reference 16) requires a minimum separation between the infiltrating surface and seasonally high mounded groundwater of at least 10 feet for infiltration basins and trenches and at least 5 feet for bioretention, planters and permeable pavement. Due to these reasons, it is our opinion that on-site subsurface infiltration of storm water is not considered feasible.

4.14 UTILITIES

Where utilities enter the pile supported structure, we recommend flexible connections be considered to help mitigate differential settlement should liquefaction settlement occur from a design level earthquake. It may also be helpful to install vaults around the connection points to allow for future access to the locations for monitoring and repairing the utilities.

4.15 PAVED AREAS

Preliminary pavement design has been based on an R-value of 20. The California Division of Highways Design Method was used for design of the recommended preliminary pavement sections. Final pavement design should be based on R-value testing performed near the conclusion of rough grading. The following pavement sections are recommended for planning purposes only.

Preliminary Pavement Sections

PAVEMENT AREA	TRAFFIC INDEX	SECTION THICKNESS (inches)	
		Asphalt Concrete	**Aggregate Base Course**
Auto Parking	4	3	5
Circulation Drives	5	3	8
Truck Drives	6	3	11
		Portland Cement Concrete	**Aggregate Base Course**
Auto Parking	4	6.5	4
Circulation Drives	5	7	4
Truck Drives	6	7	4

If vehicular pavers are to be used for the project, the paver and leveling sand may be supported on the thickness of aggregate base shown above for the appropriate traffic index. The pavement subgrade underlying the aggregate base or concrete should be properly prepared and compacted in accordance with the recommendations outlined under "Subgrade Preparation".

The portland cement concrete used for paving should have a modulus of rupture of at least 550 psi (equivalent to an approximate compressive strength of 3,700 psi) at the time the pavement is subjected to truck traffic.

The pavement base course (as well as the top 12 inches of the subgrade soils) should be compacted to at least 95 percent of the maximum dry density (ASTM D-1557). Aggregate base should conform to the requirements of Section 26 of the California Department of Transportation Standard Specifications for Class II aggregate base (three-quarter inch maximum) or Section 200-2 of the Standard Specifications for Public Works Construction (Green Book) for untreated base materials, excluding processed miscellaneous base.

The above recommendations are based on the assumption that the base course and compacted subgrade will be properly drained. The design of paved areas should incorporate measures to prevent moisture build-up within the base course which can otherwise lead to premature pavement failure. For example, curbing adjacent to landscaped areas should be deep enough to act as a barrier to infiltration of irrigation water into the adjacent base course.

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5.0 LIMITATIONS

The report, exploration logs, and other materials resulting from GPI's efforts were prepared exclusively for use by Bella Terra Associates, LLC, and their consultants in designing the proposed development. The report is not intended to be suitable for reuse on extensions or modifications of the project or for use on any project other than the currently proposed development as it may not contain sufficient or appropriate information for such uses. If this report or portions of this report are provided to contractors or included in specifications, it should be understood that they are provided for information only.

Soil deposits may vary in type, strength, and many other important properties between points of exploration due to non-uniformity of the geologic formations or to man-made cut and fill operations. While we cannot evaluate the consistency of the properties of materials in areas not explored, the conclusions drawn in this report are based on the assumption that the data obtained in the field and laboratory are reasonably representative of field conditions and are conducive to interpolation and extrapolation.

Furthermore, our recommendations were developed with the assumption that a proper level of field observation and construction review will be provided during grading, excavation, and foundation construction by GPI. If field conditions during construction appear to be different than is indicated in this report, we should be notified immediately so that we may assess the impact of such conditions on our recommendations. If construction phase services are performed by others they must accept full responsibility for all geotechnical aspects of the project including this report.

Our investigation and evaluations were performed using generally accepted engineering approaches and principles available at this time and the degree of care and skill ordinarily exercised under similar circumstances by reputable Geotechnical Engineers practicing in this area. No other representation, either expressed or implied, is included or intended in our report.

Respectfully submitted,
Geotechnical Professionals Inc.

Donald A. Cords, G.E.
Principal

James E. Harris, G.E.
Principal

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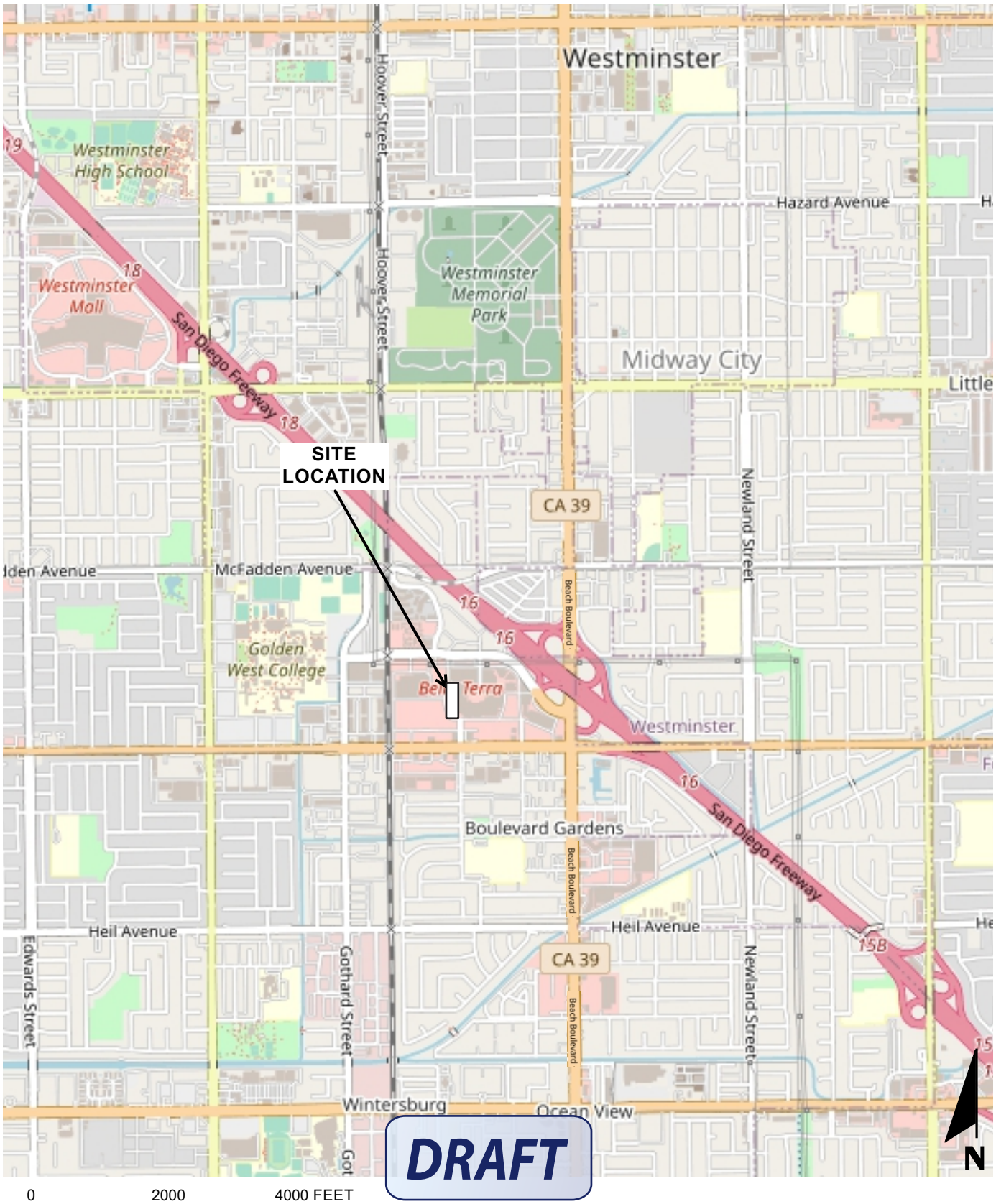
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BASE MAP REPRODUCED FROM OPEN STREET MAPS © CALTOPO

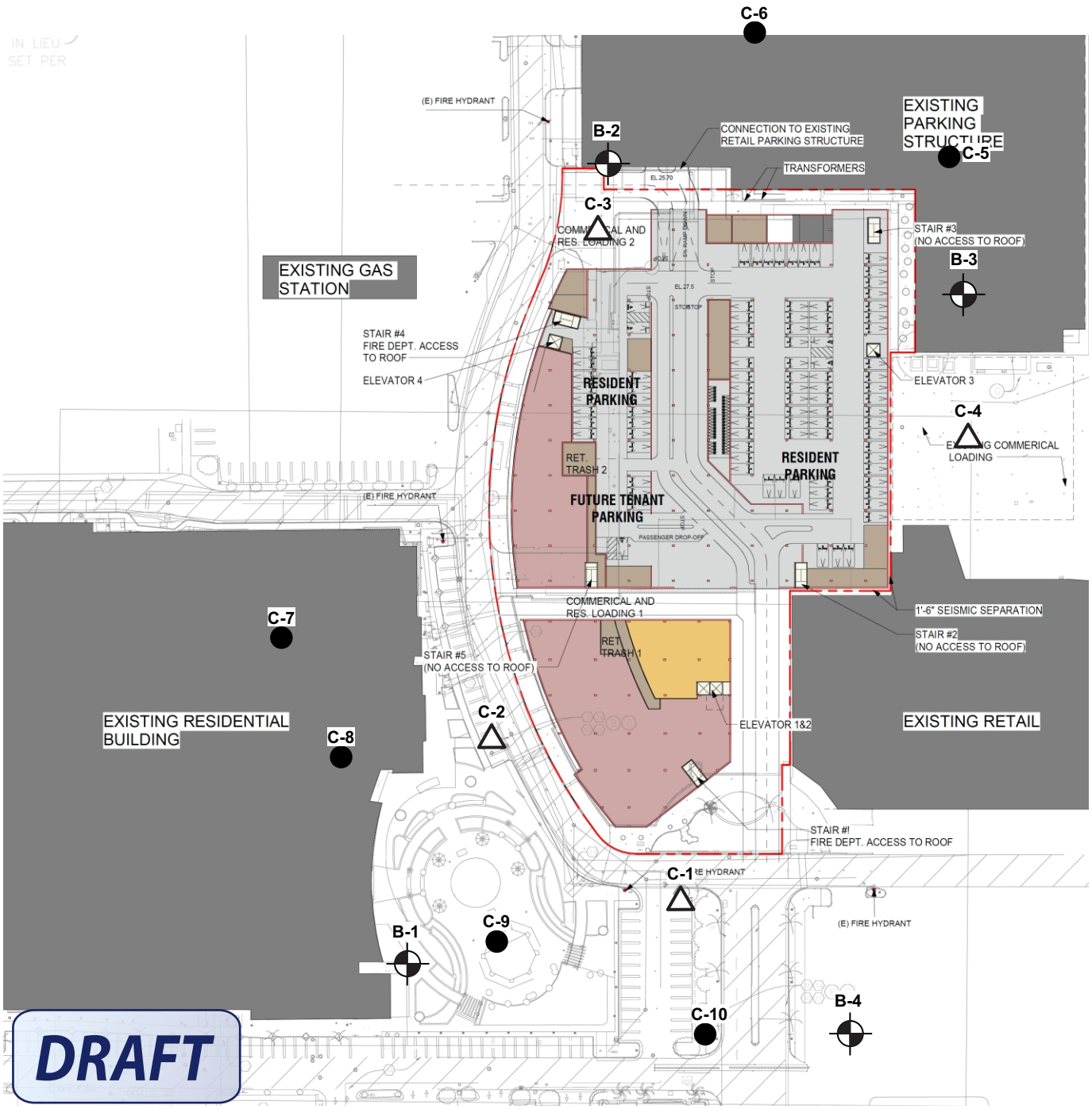
GPI GEOTECHNICAL PROFESSIONALS, INC.

BELLA TERRA RESIDENTIAL

GPI PROJECT NO.: 3015.I | SCALE: 1" = 2000'

SITE LOCATION MAP

FIGURE 1



EXPLANATION

C-4 △ APPROXIMATE LOCATION AND NUMBER OF CONE PENETRATION TEST (CURRENT)

B-4 ⊕ APPROXIMATE LOCATION AND NUMBER OF PRIOR EXPLORATORY BORINGS (1588.I)

C-10 ● APPROXIMATE LOCATION AND NUMBER OF PRIOR CONE PENETRATION TEST(1588.I)

N

0 120 240 FEET

BASE PLAN REPRODUCED FROM SITE PLAN SHEET A-0.4
PROVIDED BY TCA ARCHITECTS DATED 12-08-2020

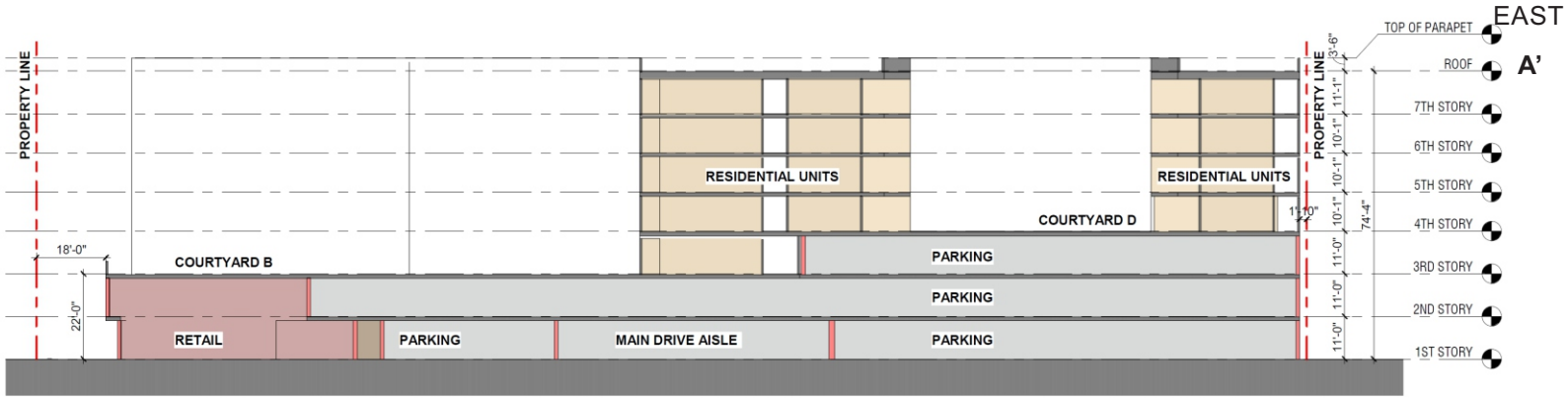
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BELLA TERRA RESIDENTIAL

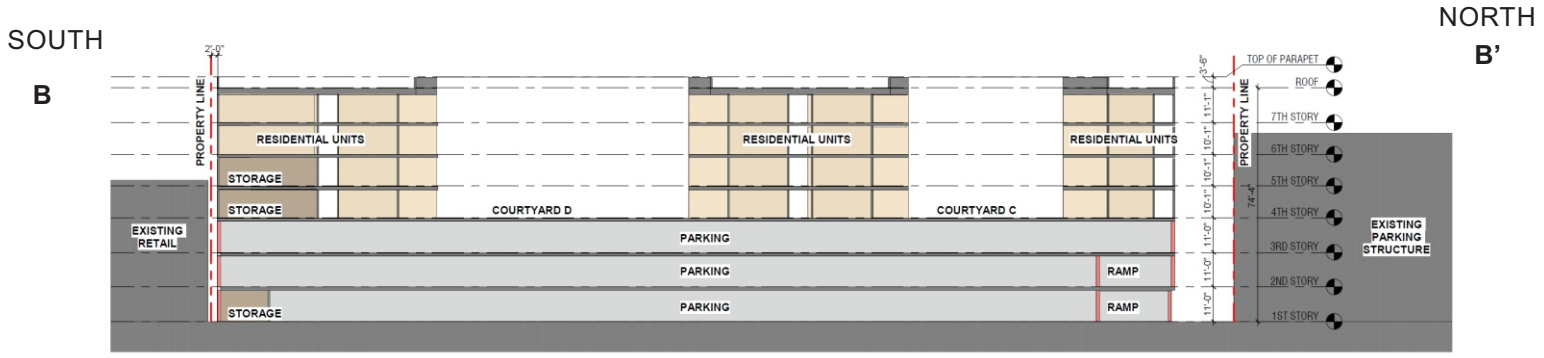
GPI PROJECT NO.: 3015.I | SCALE: 1" = 120'

SITE PLAN

FIGURE 2



SECTION A-A'



1 SECTION B
1" = 20'-0"

SECTION B-B'

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GEOTECHNICAL PROFESSIONALS, INC.

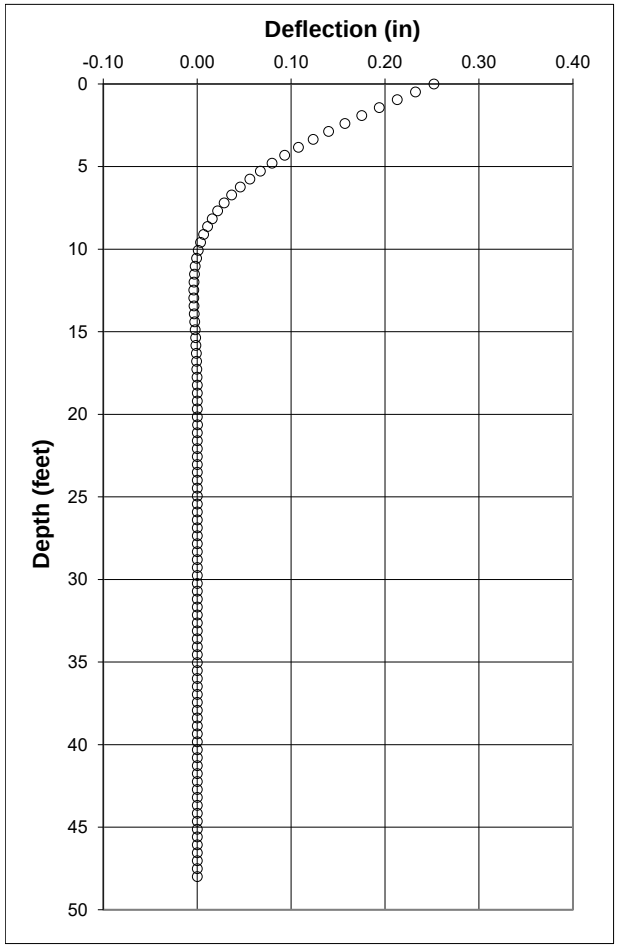
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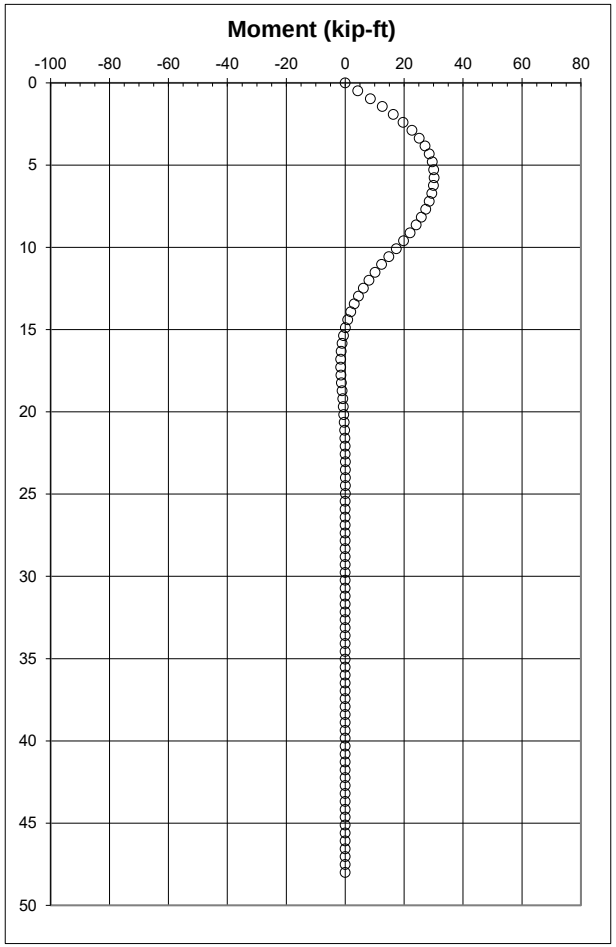
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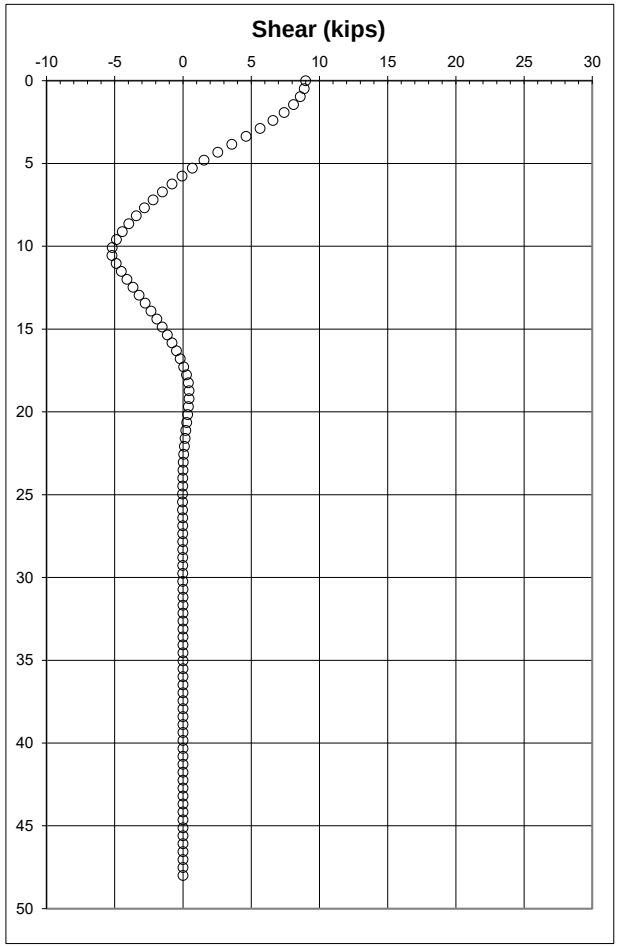
SCALE: 1" = 50'

BUILDING CROSS SECTIONS

FIGURE 3





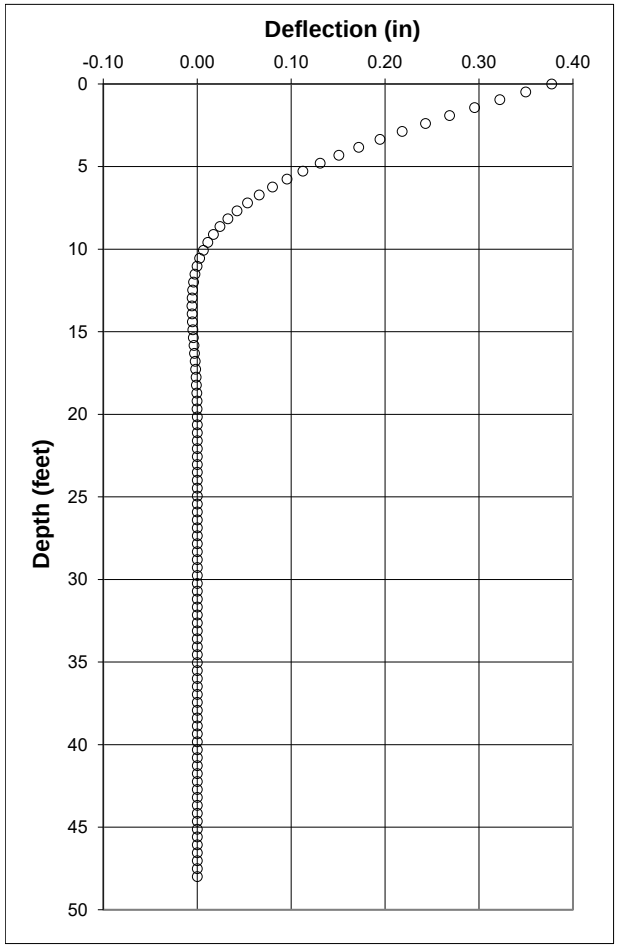


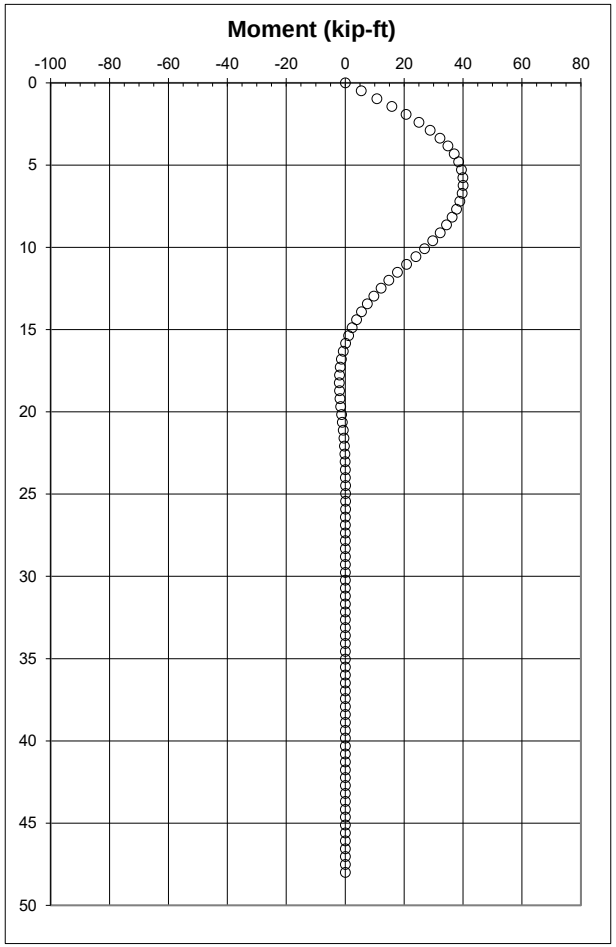
FREE HEAD LATERAL PILE ANALYSIS
16" APGD PILE - 1/4 INCH DEFLECTION; 9 KIPS
BELLA TERRA RESIDENTIAL, HUNTINGTON BEACH, CALIFORNIA
GPI Project No.: 3015.I

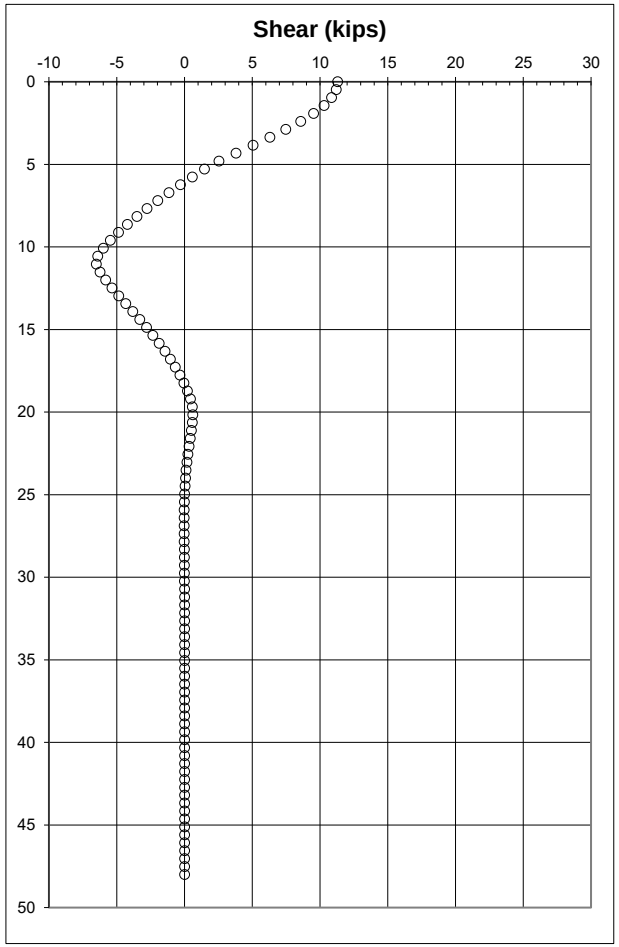
GEOTECHNICAL PROFESSIONALS, INC.

FIGURE 4A

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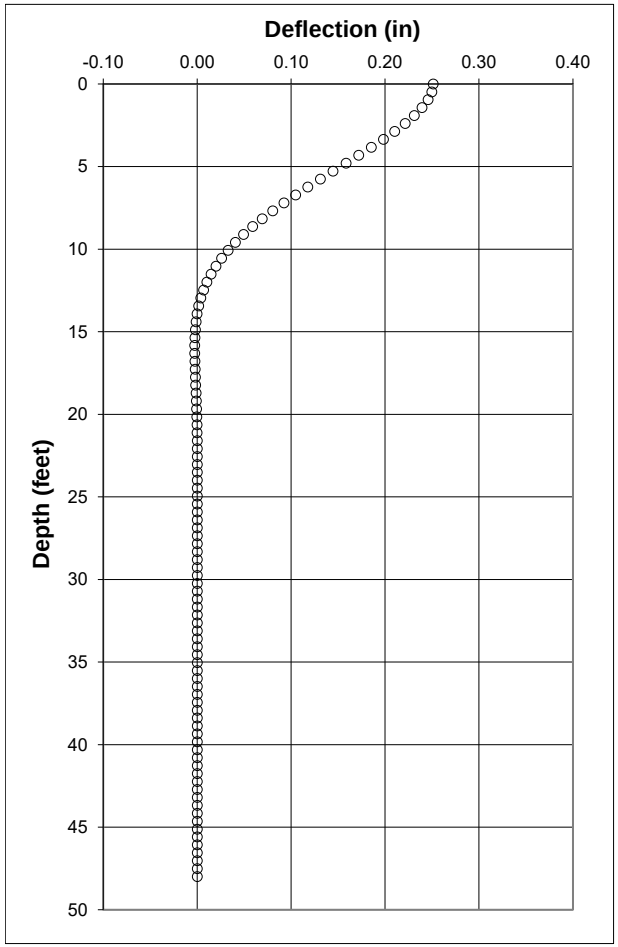


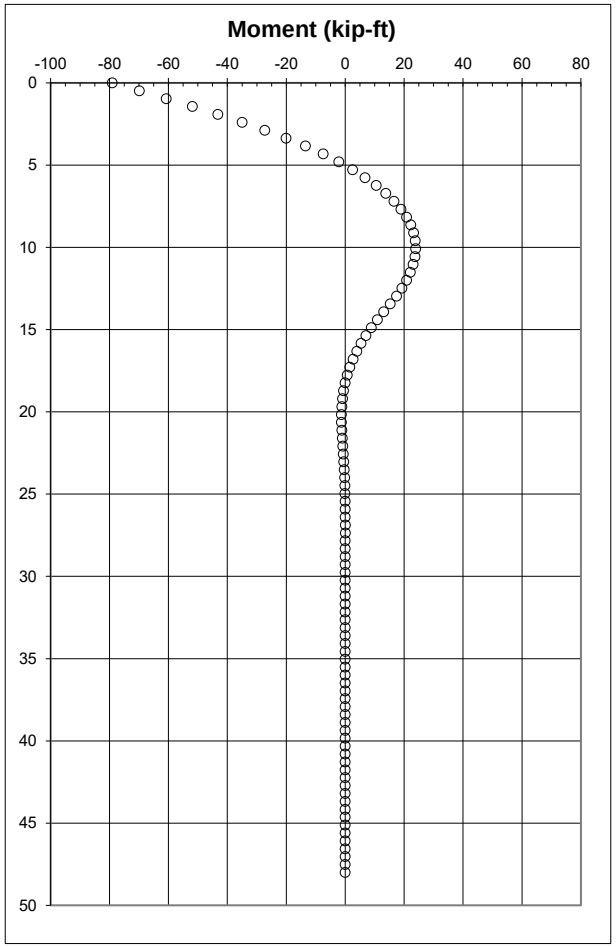
FREE HEAD LATERAL PILE ANALYSIS
16" APGD PILE - 3/8 INCH DEFLECTION; 11.5 KIPS
BELLA TERRA RESIDENTIAL, HUNTINGTON BEACH, CALIFORNIA
GPI Project No.: 3015.I

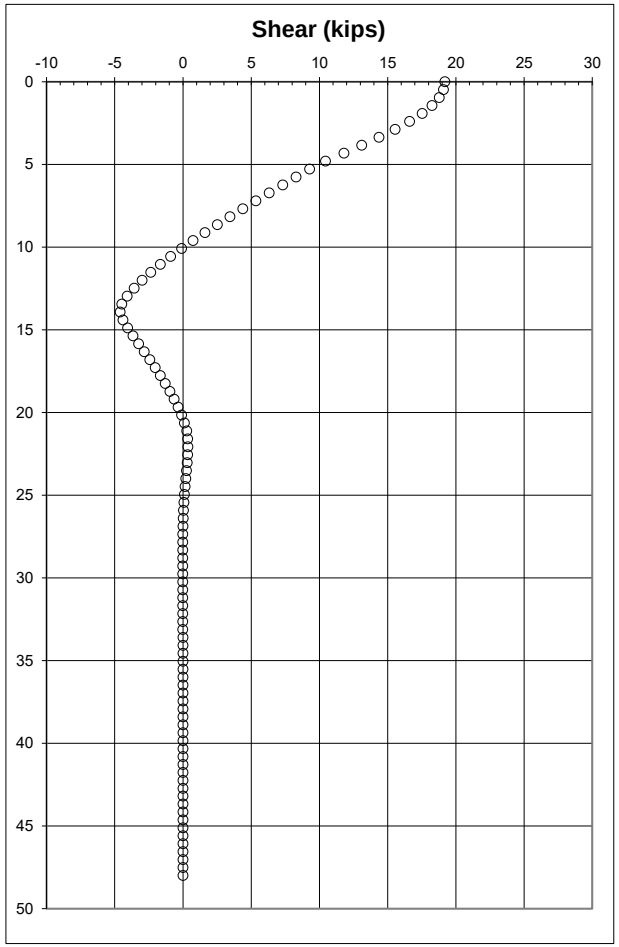
GEOTECHNICAL PROFESSIONALS, INC.

FIGURE 4B

DRAFT





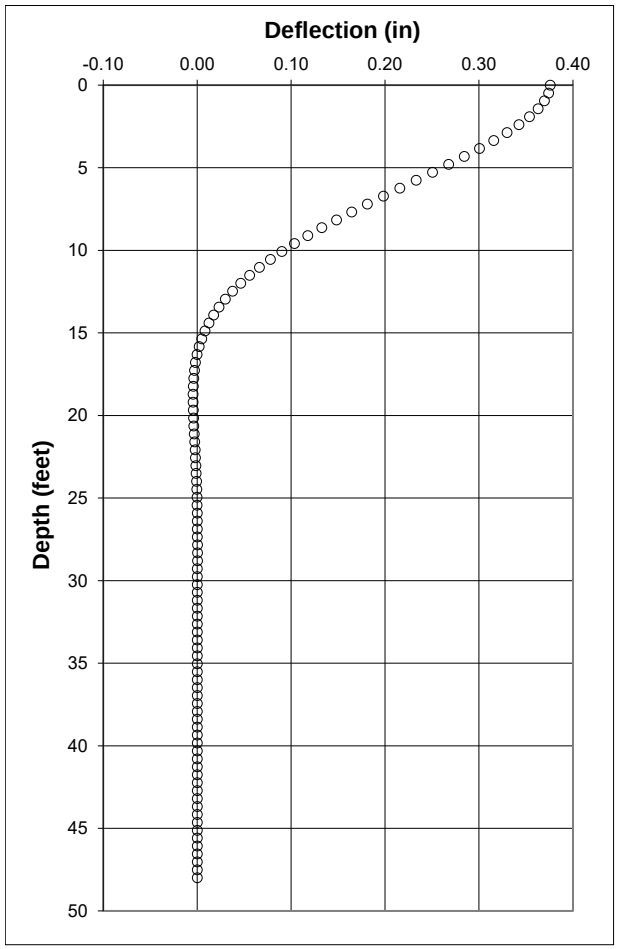


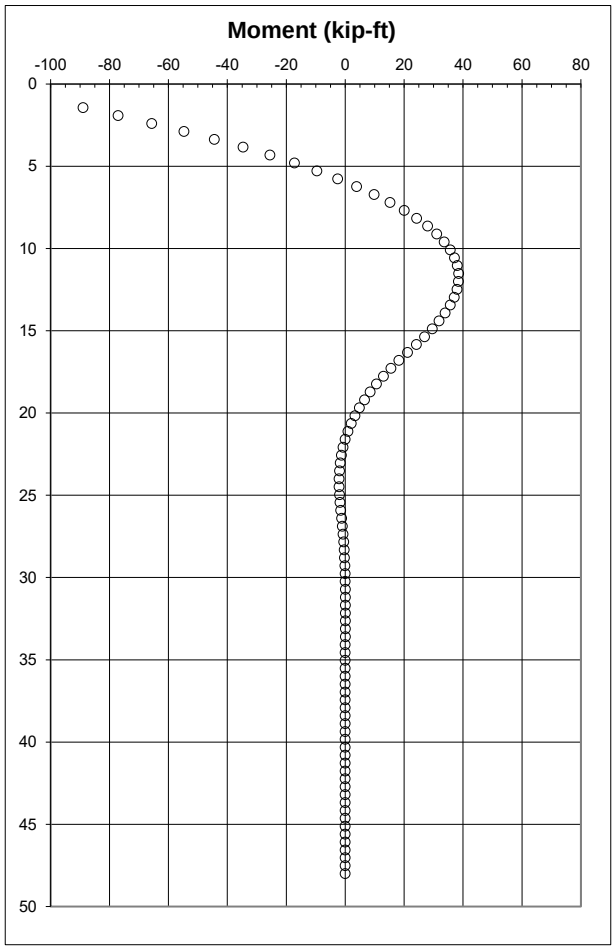
FIXED HEAD LATERAL PILE ANALYSIS
16" APGD PILE - 1/4 INCH DEFLECTION; 19 KIPS
BELLA TERRA RESIDENTIAL, HUNTINGTON BEACH, CALIFORNIA
GPI Project No.: 3015.I

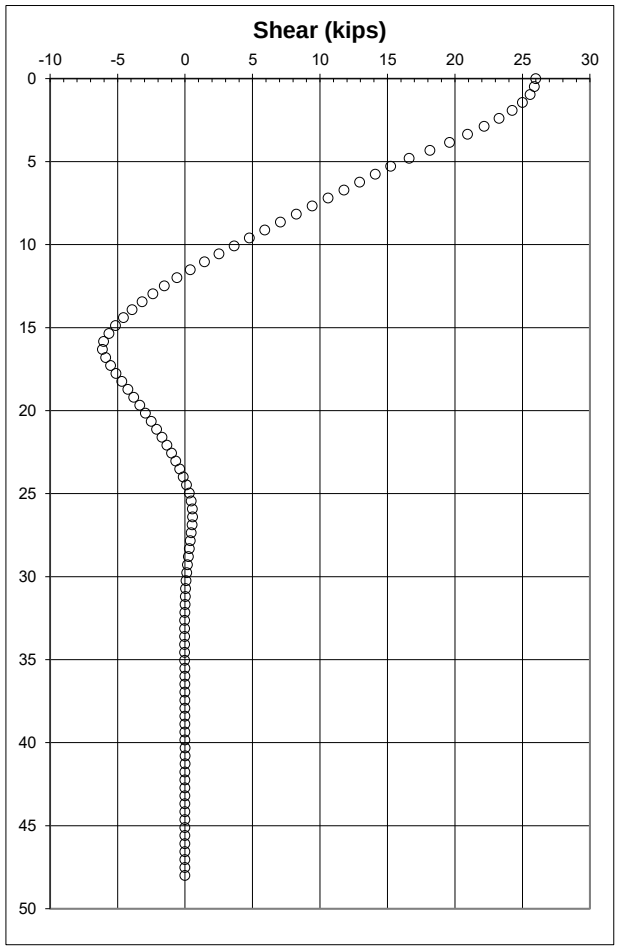
GEOTECHNICAL PROFESSIONALS, INC.

FIGURE 5A

DRAFT







FIXED HEAD LATERAL PILE ANALYSIS
16" APGD PILE - 3/8 INCH DEFLECTION; 24 KIPS
BELLA TERRA RESIDENTIAL, HUNTINGTON BEACH, CALIFORNIA
GPI Project No.: 3015.I

GEOTECHNICAL PROFESSIONALS, INC.

FIGURE 5B

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APPENDIX A

APPENDIX A

CONE PENETRATION TESTS

The subsurface conditions were investigated by performing ten Cone Penetration Tests (CPT's) at the site. These soundings were advanced to depths ranging from approximately 50 to 80 feet below existing grades. The locations of the CPT's are shown on the Site Plan, Figure 2.

Six CPT's (C-5 through C-10) were used from our previous investigations at the site (References 1 to 5). The CPT's were performed in 2000, 2001, and 2010 as part of the overall site redevelopment for Bella Terra. We have modified the numerical designation of the CPT's from the previous reports. Notes on the logs provide the original designation of the CPT's as shown in our 2002 updated geotechnical investigation (Reference 1) and 2010 geotechnical investigation (Reference 5).

The Cone Penetration Test consists of pushing a cone-tipped probe into the soil deposit while simultaneously recording the cone tip resistance and side friction resistance of the soil to penetration (refer to Figure A-1). The CPT's described in this report were conducted in general accordance with ASTM specifications (ASTM D 5778) using an electric cone penetrometer.

The CPT equipment consists of a cone assembly mounted at the end of a series of hollow sounding rods. A set of hydraulic rams is used to push the cone and rods into the soil while a continuous record of cone and friction resistance versus depth is obtained in both analog and digital form at the ground surface. A specially designed truck is used to transport and house the test equipment and to provide a reaction to the thrust of the hydraulic rams.

Data obtained during a CPT consists of continuous stratigraphic information with close vertical resolution. Stratigraphic interpretation is based on relationships between cone tip resistance and friction resistance. The calculated friction ratio (CPT friction sleeve resistance divided by cone tip resistance) is used as an indicator of soil type. Granular soils typically have low friction ratios and high cone resistance, while cohesive or organic soils have high friction ratios and low cone resistance. These stratigraphic material categories form the basis for all subsequent calculations which utilize the CPT data.

Computer plots of the reduced CPT data acquired for this investigation are presented in Figures A-2 to A-11 of this appendix. The field testing and computer processing for the current investigation was performed by both Fugro Geosciences and Kehoe Engineering and Testing under subcontract to Geotechnical Professionals Inc. (GPI). The interpreted soil descriptions were prepared by GPI.

A seismic cone penetration test provided shear wave velocity measurements of the soil profile. A standard cone penetrometer is equipped with two sets of geophones located approximately 3 feet apart on the cone penetrometer. At 5 foot intervals, a shear wave source is activated at the ground surface using an air-actuated hammer. A seismograph measures the travel time of the shear wave detected at each set of geophones. The time difference provides the velocity of the shear wave in the layer between the two geophone sets.

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Seismic cone penetration tests was performed at CPT C-4. The cone penetration test at location C-3 refused at approximately 80 feet and the subsequent data was used in order to estimate the average shear wave velocity for the upper 100 feet of soil profile. Table A-1 provides the shear wave velocity from the surface and the interval of soil between the geophones.

The CPT locations were laid out in the field by measuring from existing site features. Ground surface elevations at the CPT locations were estimated from the topography on a Conceptual Grading Plan (provided October 7, 2008) prepared by Hunsaker & Associates.

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TABLE A-1
SEISMIC SHEAR WAVE VELOCITY MEASUREMENTS

Geotechnical Professionals Inc.
Bella Terra Residential
Huntington Beach, CA

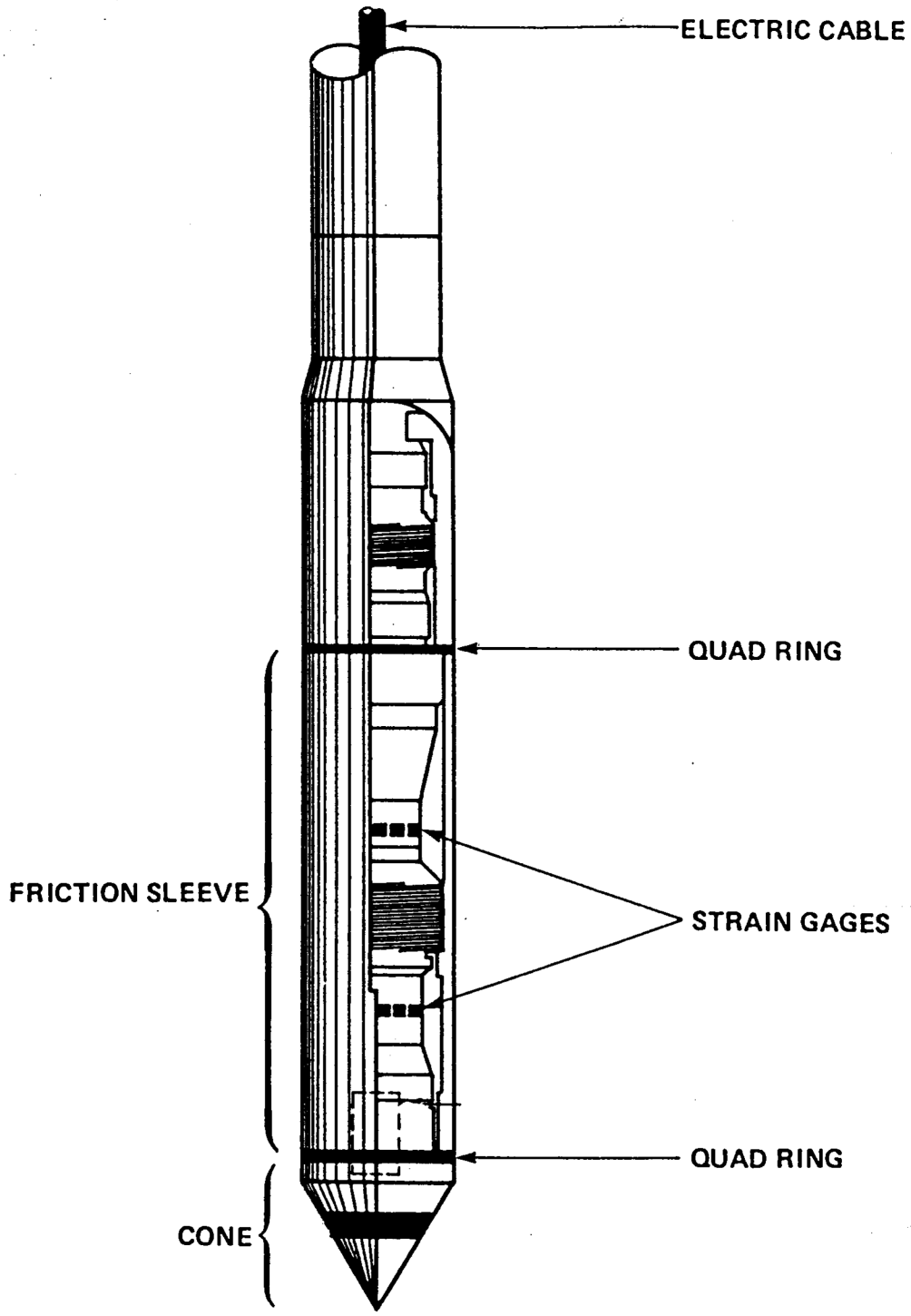
CPT Shear Wave Measurements

Location	Tip Depth (ft)	Geophone Depth (ft)	Travel Distance (ft)	S-Wave Arrival (msec)	S-Wave Velocity from Surface (ft/sec)	Interval S-Wave Velocity (ft/sec)
CPT-4	5.15	4.15	4.61	4.48	1028	
	10.10	9.10	9.32	16.36	570	396
	15.09	14.09	14.23	25.88	550	516
	20.11	19.11	19.21	39.28	489	372
	25.07	24.07	24.15	51.48	469	405
	30.05	29.05	29.12	62.82	464	438
	35.07	34.07	34.13	74.20	460	440
	40.03	39.03	39.08	83.84	466	514
	45.01	44.01	44.06	91.64	481	638
	50.07	49.07	49.11	97.60	503	848
	55.05	54.05	54.09	103.84	521	797
	60.07	59.07	59.10	110.14	537	796
	65.09	64.09	64.12	115.00	558	1032
	70.05	69.05	69.08	119.26	579	1164
	75.03	74.03	74.06	125.58	590	788
	80.09	79.09	79.12	129.74	610	1216
	100.00				Assumed value:	1216

Average Shear Wave Velocity in upper 100 feet: 813 ft/sec

Shear Wave Source Offset - 2 ft

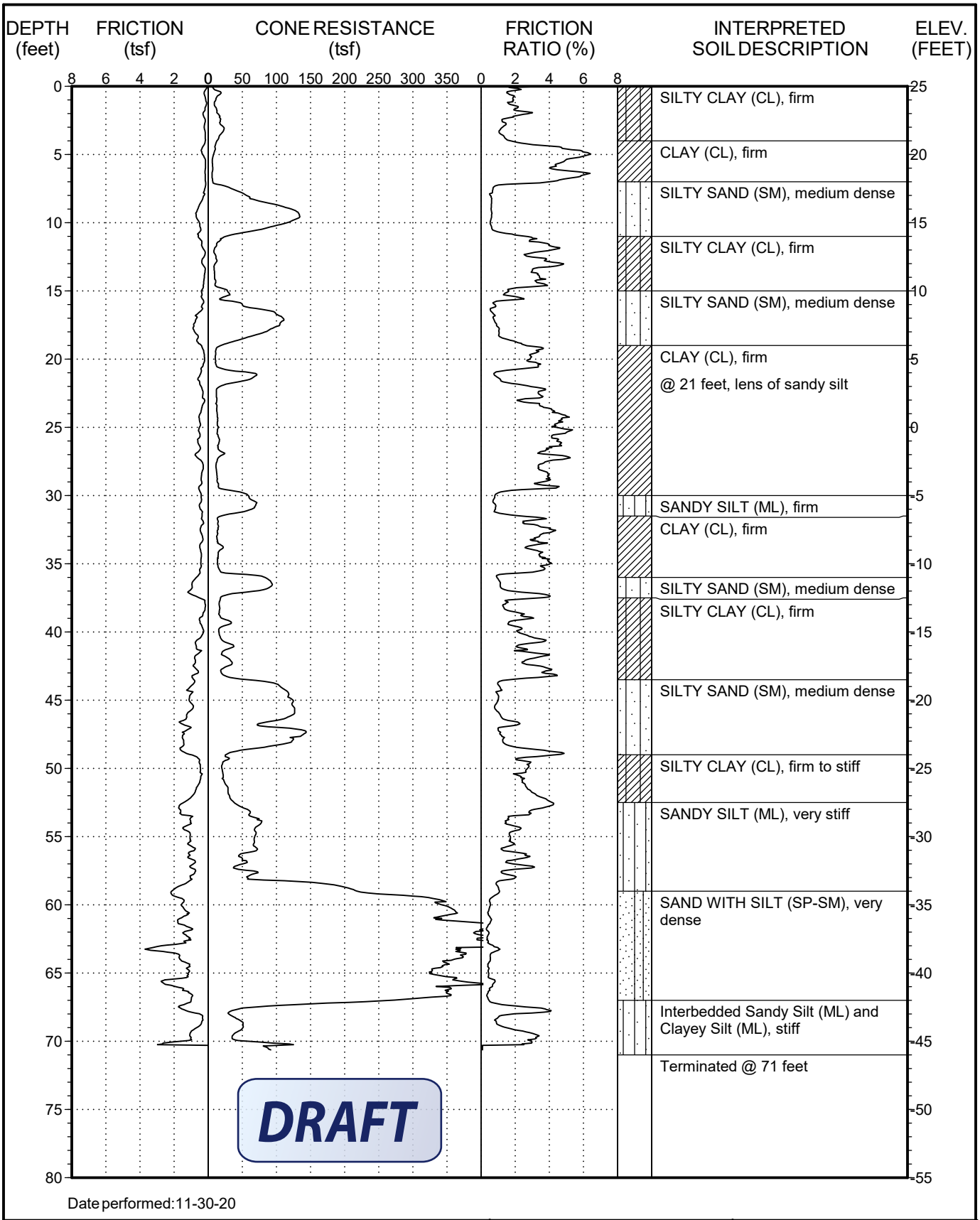
S-Wave Velocity from Surface = Travel Distance/S-Wave Arrival
Interval S-Wave Velocity = (Travel Dist2-Travel Dist1)/(Time2-Time1)



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CONE PENET O ETER

FIGURE



Date performed:11-30-20

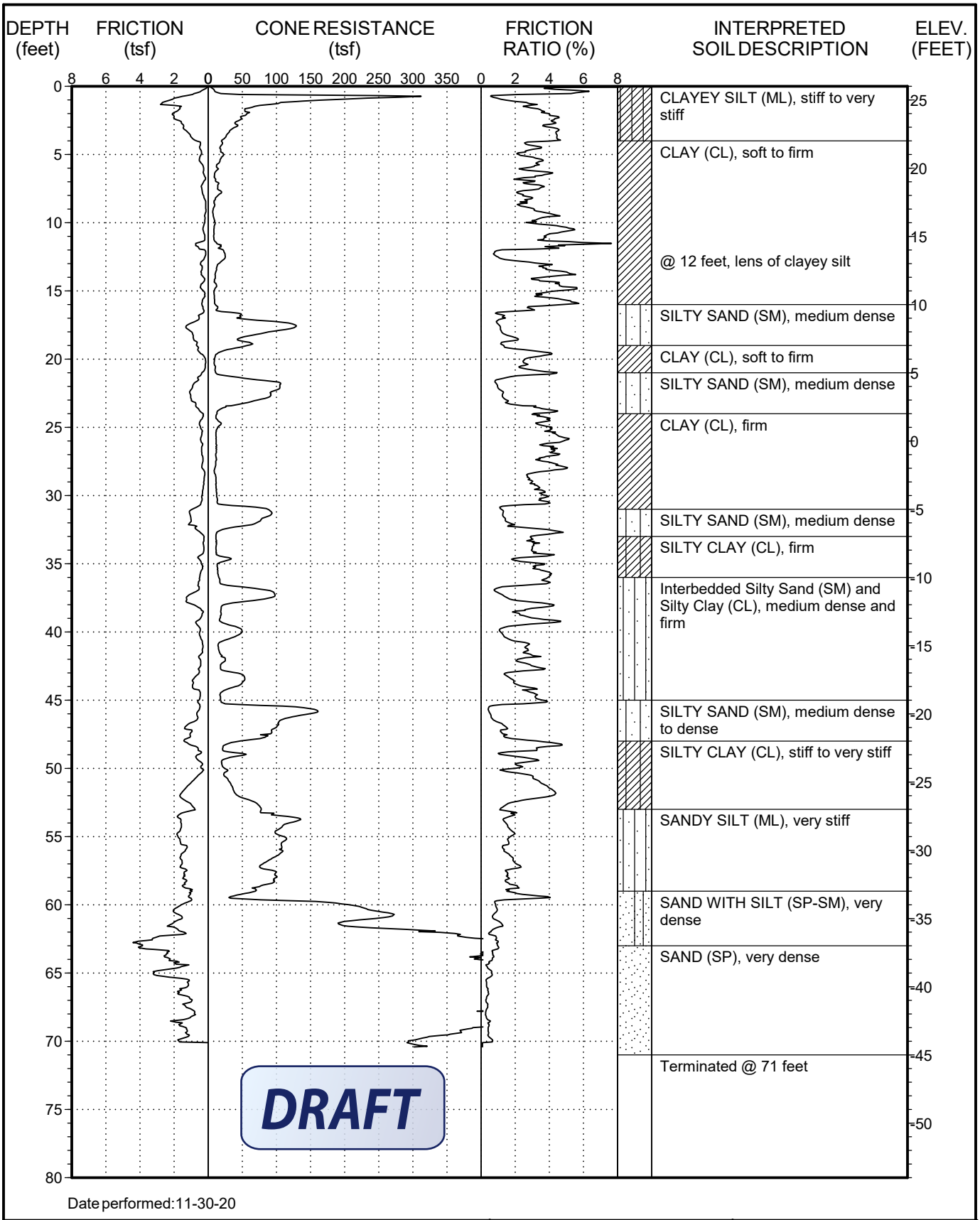
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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LOG OF CPT NO. C-1

FIGURE A-2



Date performed:11-30-20

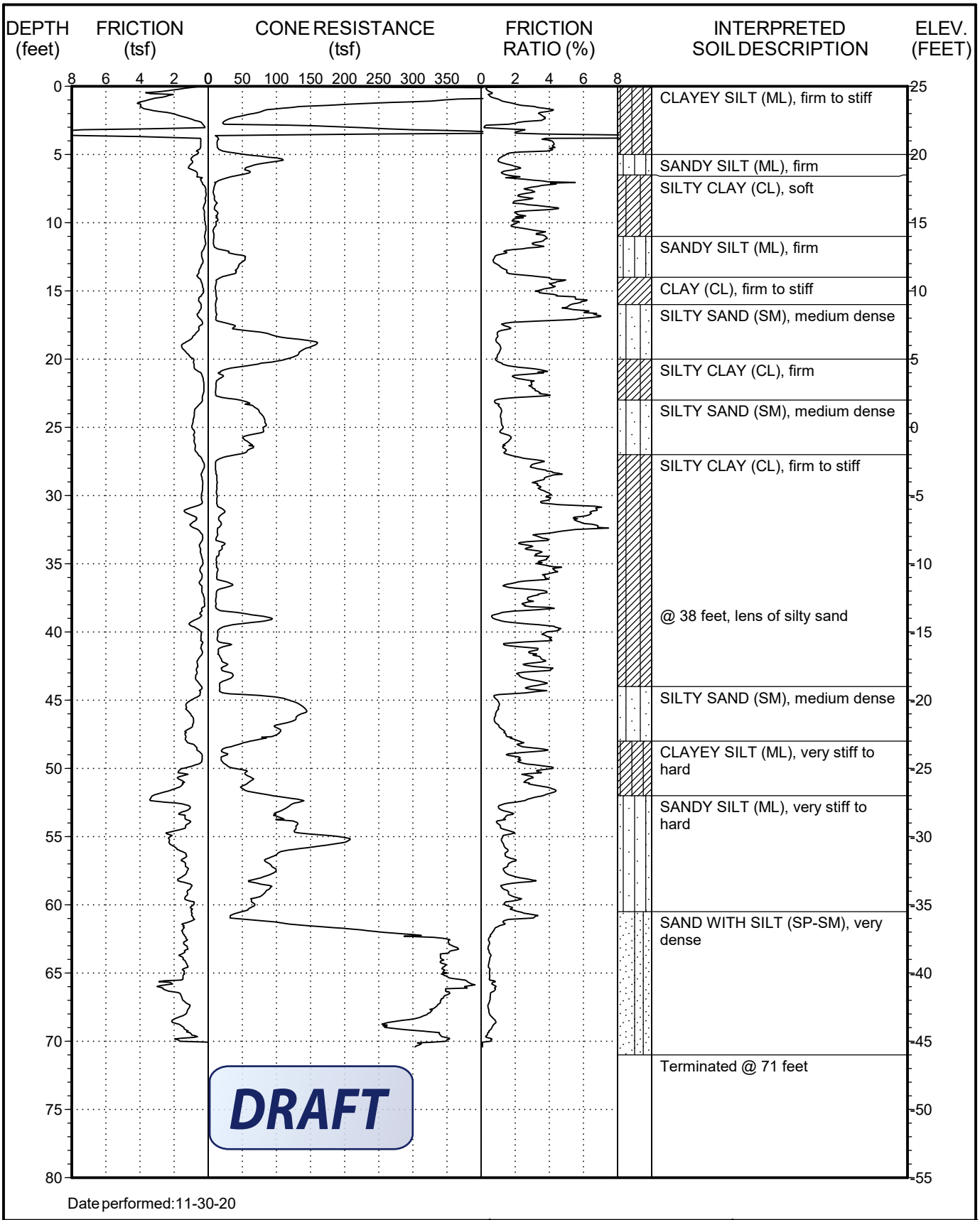
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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BELLA TERRA APARTMENTS

LOG OF CPT NO. C-2

FIGURE A-3



Date performed:11-30-20

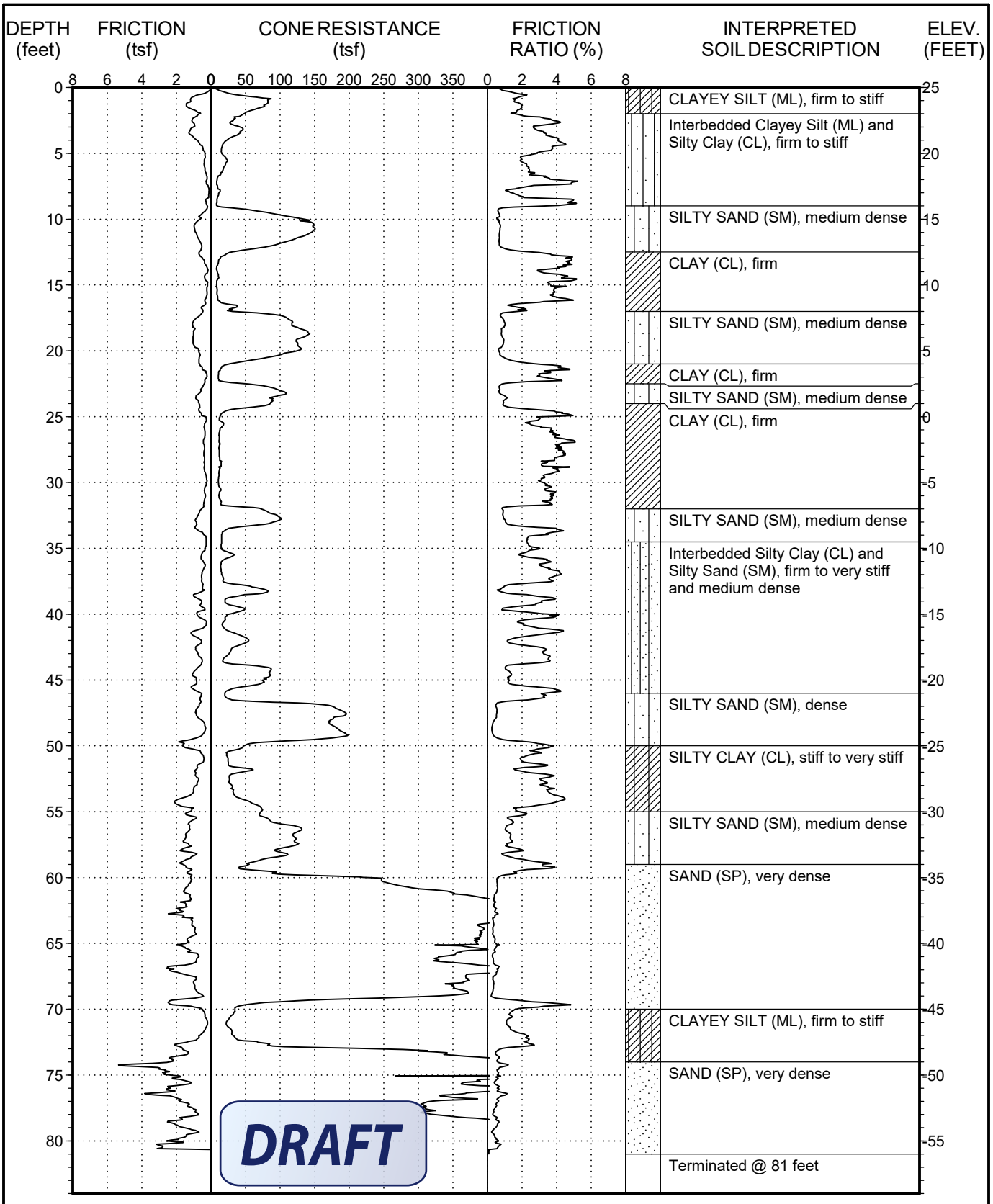
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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BELLA TERRA APARTMENTS

LOG OF CPT NO. C-3

FIGURE A-4



Date performed:11-30-20

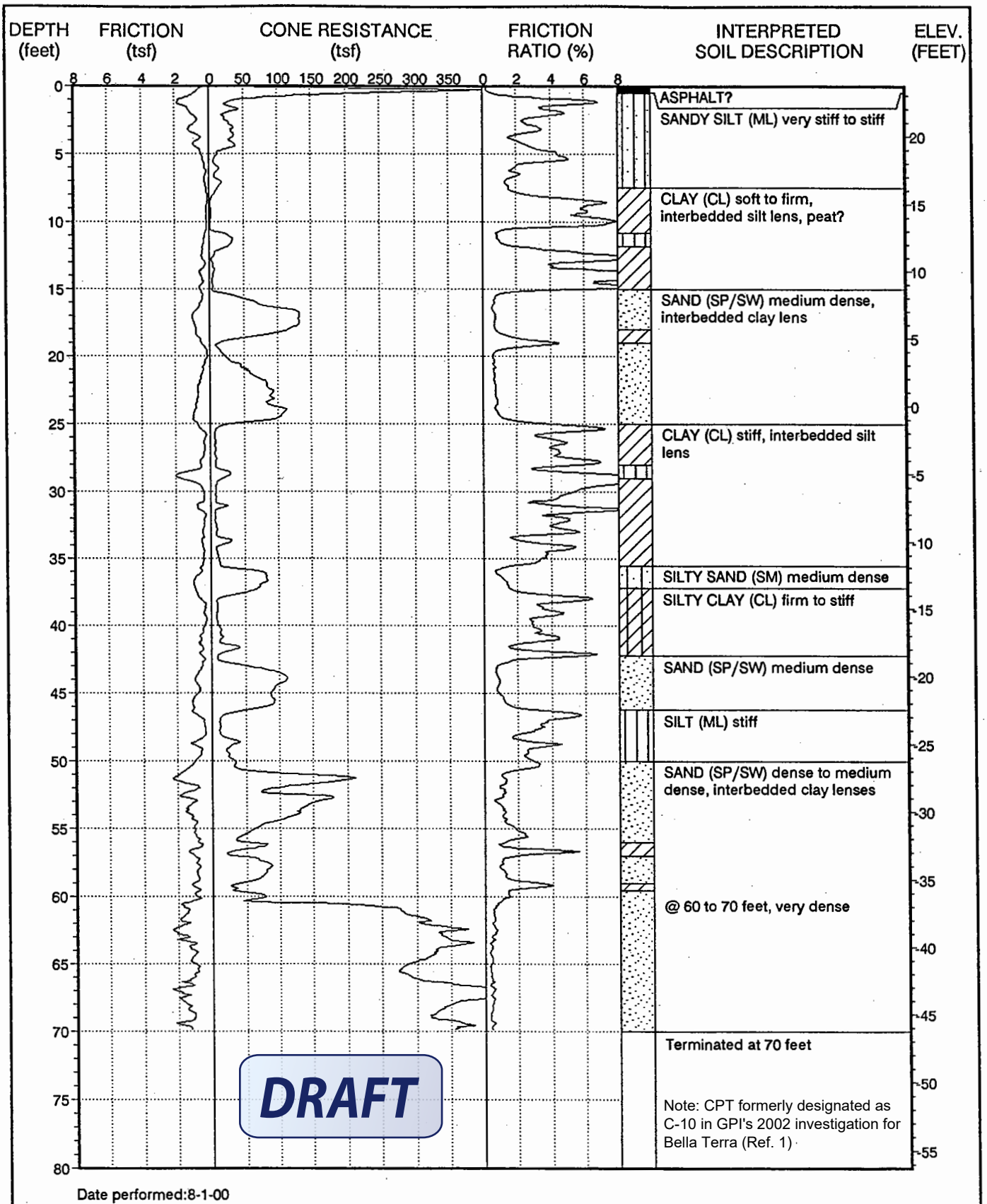
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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BELLA TERRA APARTMENTS

LOG OF CPT NO. C-4

FIGURE A-5



Date performed:8-1-00

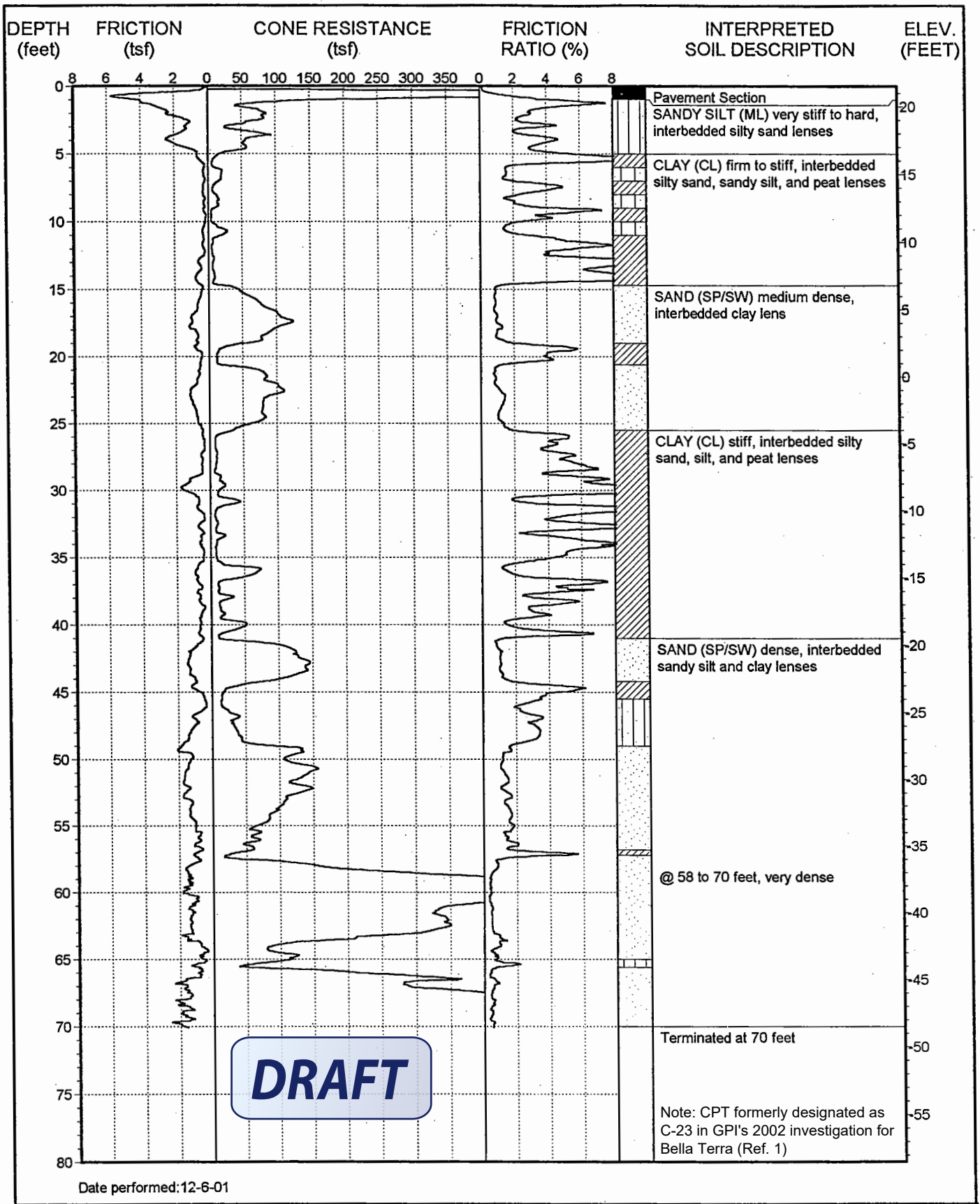
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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PROJECT NO.: 3015.I
BELLA TERRA RESIDENTIAL

LOG OF CPT NO. C-5

FIGURE A-6



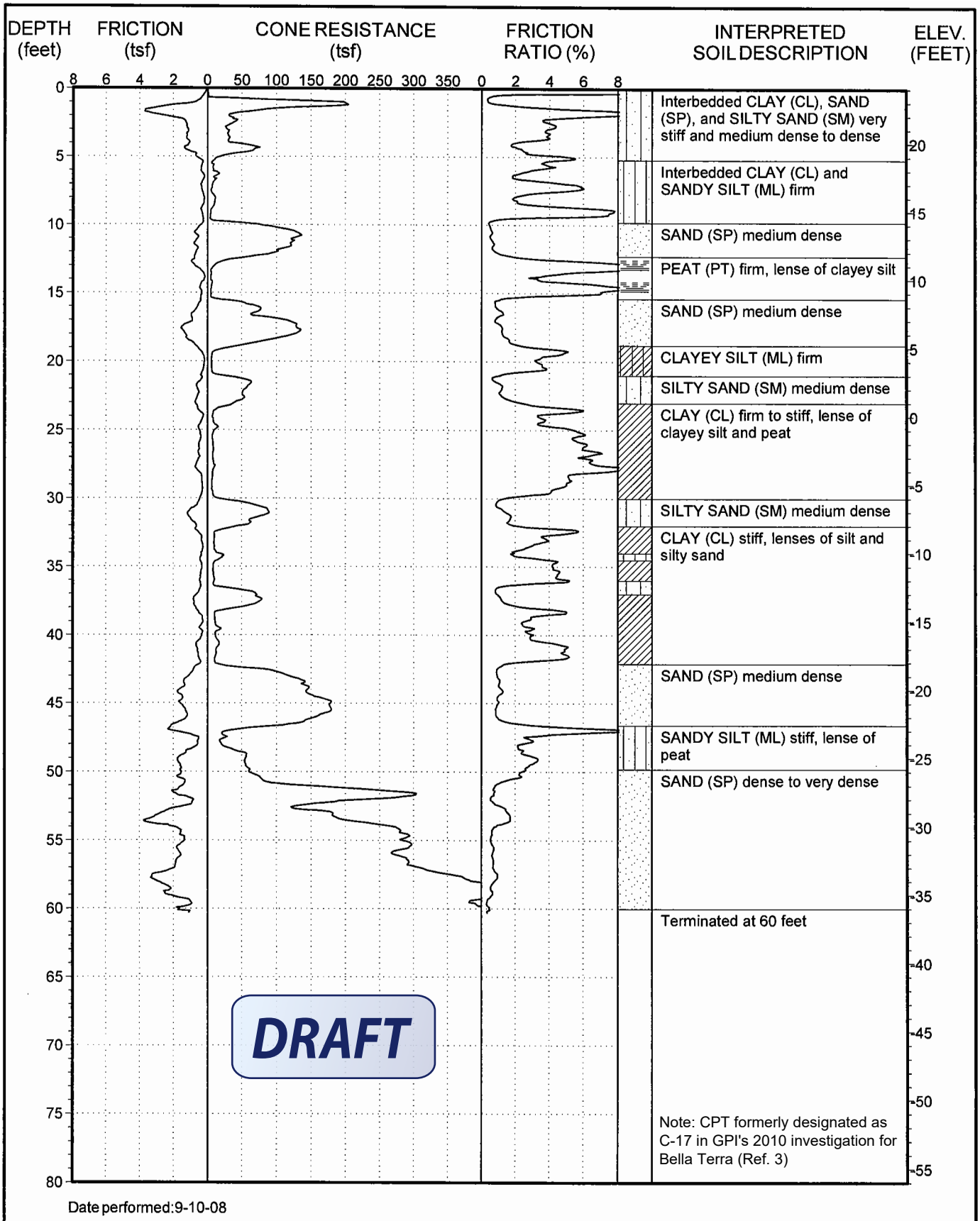
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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BELLA TERRA RESIDENTIAL

LOG OF CPT NO. C-6

FIGURE A-7



Date performed:9-10-08

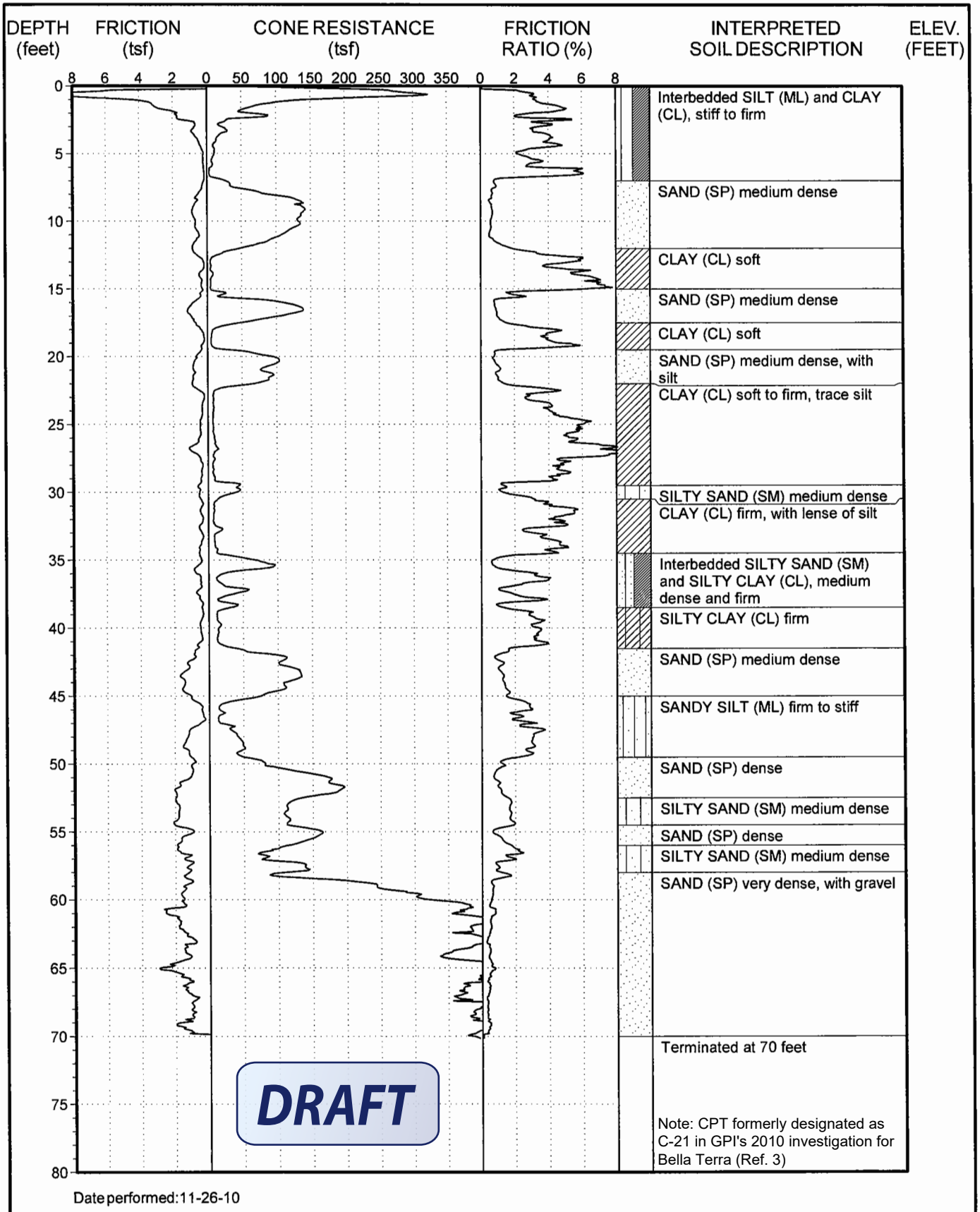
This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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BELLA TERRA RESIDENTIAL

LOG OF CPT NO. C-7

FIGURE A-8



Date performed: 11-26-10

This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

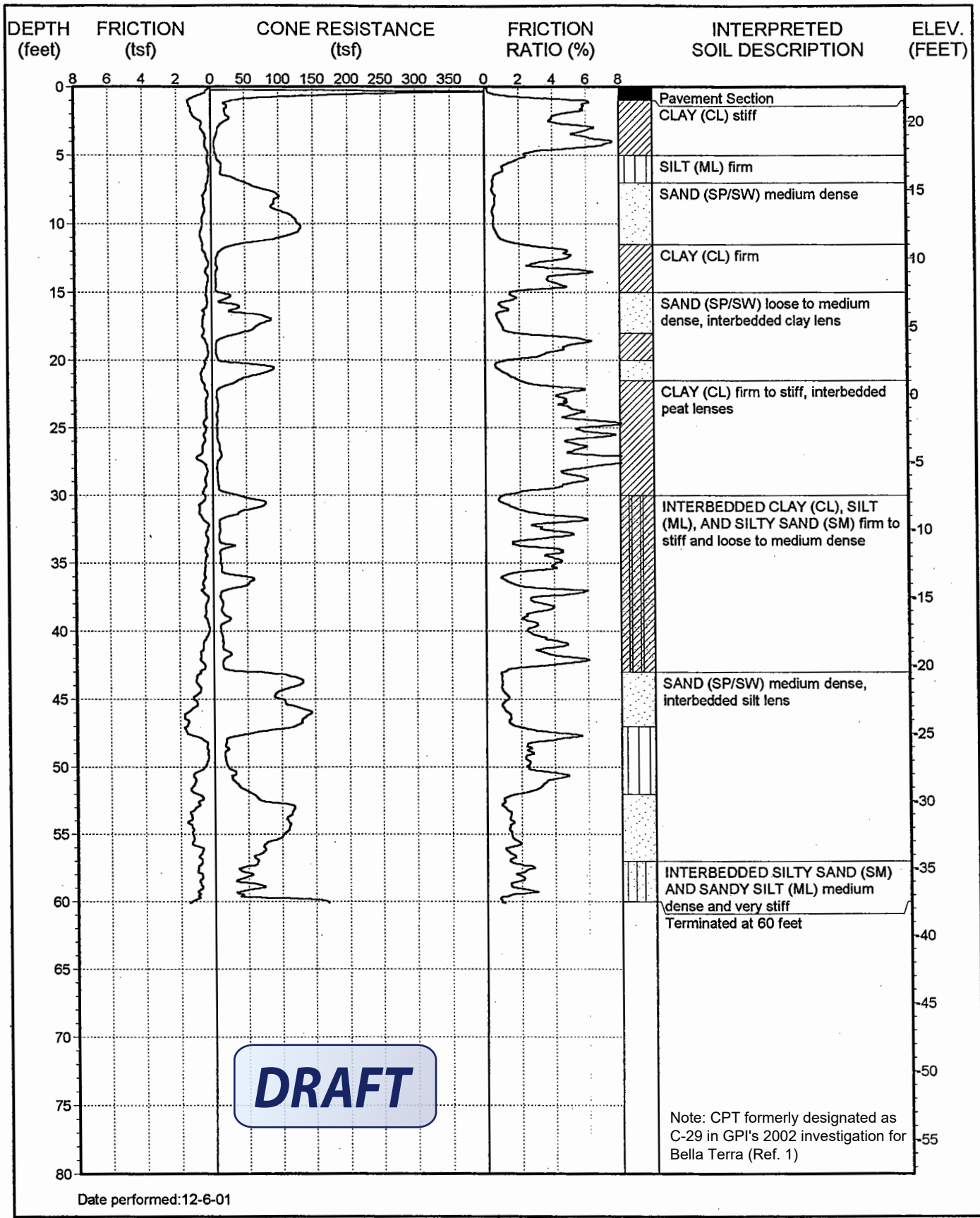
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LOG OF CPT NO. C-8

FIGURE A-9



Date performed:12-6-01

This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

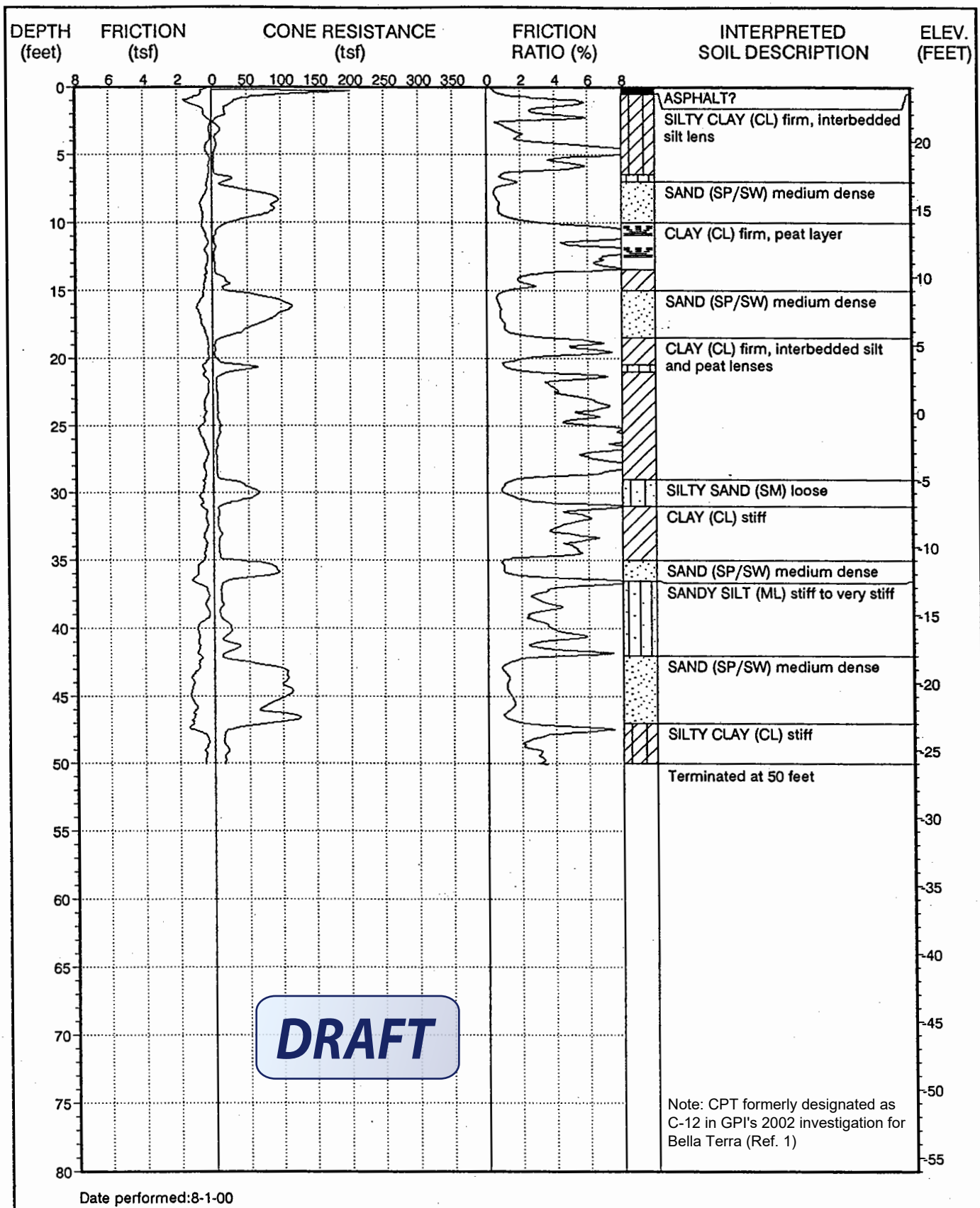
GPI

PROJECT NO.: 3015.I

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LOG OF CPT NO.C-9

FIGURE A-10



Date performed:8-1-00

This summary applies only at the location of this cone penetration test and at the time of the exploration. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The interpreted soil description is derived from the friction ratio and cone resistance and is a simplification of actual conditions encountered.

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LOG OF CPT NO. C-10

FIGURE A-11

APPENDIX B

APPENDIX B

EXPLORATORY BORINGS

The subsurface conditions at the site were investigated by drilling and sampling four exploratory borings. The borings were advanced to depths ranging from 31 to 51½ feet below the existing ground surface. The locations of the explorations are shown on the Site Plan, Figure 2.

These borings were performed during our previous investigations at the site (References 1 to 4). The borings were performed in 2000 and 2001 as part of an overall site redevelopment for Bella Terra. We have modified the numerical designation of the borings from the previous reports. Notes on the log of borings provide the original designation of the borings as shown in our 2002 updated geotechnical investigation (Reference 1).

The borings were drilled using truck-mounted rotary-wash drill equipment. Relatively undisturbed samples were obtained using a brass-ring lined sampler (ASTM D 3550), or thin-walled tube sampler. The brass-rings have an inside diameter of 2.42 inches. The ring samples were driven into the soil by a 400-pound hammer dropping 18 inches. The thin-walled sampler is pushed into the subgrade using the Kelly weight. The number of blows needed to drive the sampler into the soil was recorded as the penetration resistance.

At selected locations, disturbed samples were obtained using a split-spoon sampler by means of the Standard Penetration Test (SPT, ASTM D 6066). The spoon sampler was driven into the soil by a 140-pound hammer dropping 30 inches, employing the “free-fall” hammer described above. After an initial seating drive of 6 inches, the number of blows needed to drive the sampler into the soil a depth of 12 inches was recorded as the penetration resistance.

The field explorations for the investigation were performed under the continuous technical supervision of GPI's representative, who visually inspected the site, maintained detailed logs of the borings, classified the soils encountered, and obtained relatively undisturbed samples for examination and laboratory testing. The soils encountered in the borings were classified in the field and through further examination in the laboratory in accordance with the Unified Soils Classification System. Detailed logs of the borings are presented in Figures B-1 to B-4 in this appendix.

The boring locations were laid out in the field by measuring from existing site features. Ground surface elevations at the boring locations were estimated from the topography on a Conceptual Grading Plan (provided October 7, 2008) prepared by Hunsaker & Associates.

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MOISTURE (%)	DRY DENSITY (PCF)	PENETRATION RESISTANCE (BLOWS/FOOT)	SAMPLE TYPE	DEPTH (FEET)	DESCRIPTION OF SUBSURFACE MATERIALS	ELEVATION (FEET)
				0	This summary applies only at the location of this boring and at the time of drilling. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The data presented is a simplification of actual conditions encountered.	
					5 inches AC over 5 inches AB	
29.7	86	3	D		Fill?: **SILTY CLAY (CL)** with **SAND (SP/SW)** dark grey and black, wet, soft	20
34.6	80	2	D	5	Natural: **SAND WITH SILT (SP-SM)** grey and red, wet, loose, fine to medium grained	
23.5	101	14	D		**SAND (SP)** grey, wet, medium dense	15
					@ 8 to 12 feet, light brown, medium to coarse grained	
24.2	95	8	D	10	@ 10 to 12 feet, loose	
					SILTY CLAY (CL) grey, wet, stiff, some silty sand lenses, trace fine sand	10
32.6	85	7	D	15	@ 15 1/2 feet, silty sand lens	
						5
25.3	92	8	D	20	**SILTY SAND (SM)** grey, wet, loose	
					SILTY CLAY (CL) grey, wet, soft to firm	0
40.6	79	4	D	25		
						-5
30.1	90	12	D	30	**SILTY SAND (SM)** dark grey, wet, medium dense	
					CLAY (CL) dark grey, wet, stiff	-10
25.3		11	D	35		
					DRAFT	-15

SAMPLE TYPES
- C Rock Core
- S Standard Split Spoon
- D Drive Sample
- B Bulk Sample
- T Tube Sample

DATE DRILLED: 8-2-00

EQUIPMENT USED: 6" Rotary Wash

GROUNDWATER LEVEL:(ft): 6 feet

GPI

PROJECT NO.: 3015.I

BELLA TERRA RESIDENTIAL

LOG OF BORING NO. B-1

FIGURE B-1

MOISTURE (%)	DRY DENSITY (PCF)	PENETRATION RESISTANCE (BLOWS/FOOT)	SAMPLE TYPE	DEPTH (FEET)	DESCRIPTION OF SUBSURFACE MATERIALS	ELEVATION (FEET)
					This summary applies only at the location of this boring and at the time of drilling. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The data presented is a simplification of actual conditions encountered.	
31.8	88	8	S	40	**CLAY (CL)** grey, wet, stiff	-20
				45	**SILTY SAND (SM)** dark grey, wet, medium dense, fine grained, layers of sandy silt and clayey silt	-25
37.6		18	S	50		
					Total Depth 51 1/2 feet	
					Note: Boring formerly designated as B-5 in GPI's 2002 investigation for Bella Terra (Ref. 1)	

DRAFT

SAMPLE TYPES
- C Rock Core
- S Standard Split Spoon
- D Drive Sample
- B Bulk Sample
- T Tube Sample

DATE DRILLED: 8-2-00

EQUIPMENT USED: 6" Rotary Wash

GROUNDWATER LEVEL:(ft): 6 feet

GPI

PROJECT NO.: 3015.I

BELLA TERRA RESIDENTIAL

LOG OF BORING NO. B-1

FIGURE B-1

MOISTURE (%)	DRY DENSITY (PCF)	PENETRATION RESISTANCE (BLOWS/FOOT)	SAMPLE TYPE	DEPTH (FEET)	DESCRIPTION OF SUBSURFACE MATERIALS	ELEVATION (FEET)
					This summary applies only at the location of this boring and at the time of drilling. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The data presented is a simplification of actual conditions encountered.	
			B	0	5 inches AC over 4 inches AB	
19.4	102	6	D		Fill: **SILTY SAND (SM)** light brown, moist, loose, with clay, mottled	20
23.4	100	2	D	5	Fill?: **SILTY SAND (SM)** light brown, moist, loose, w/clay, mottled	
55.7	67	2	D		Natural: **SILTY CLAY (CL)** dark grey/black, wet, soft, trace peat/organics and fine sand	15
49.9	69	2	D	10		10
211.8		3	D	15	**PEAT (PT)** dark red-brown and black, wet, soft, organic odor	
					SILTY SAND (SM) dark grey, wet, loose	5
42.0	72	6	D	20	**CLAY (CL)** dark grey, wet, firm	0
27.1	96	18	D	25	**SILTY SAND (SM)** grey, wet, medium dense	
		17	D			-5
40.2	78	5	D	30	**SILTY CLAY (CL)** dark grey, wet, stiff	
					Total Depth 31 feet	

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Note: Boring also designated as B-2 in GPI's 2002 investigation for Bella Terra (Ref. 1)

SAMPLE TYPES
- C Rock Core
- S Standard Split Spoon
- D Drive Sample
- B Bulk Sample
- T Tube Sample

DATE DRILLED: 8-3-00

EQUIPMENT USED: 6" Rotary Wash

GROUNDWATER LEVEL:(ft): 10 1/2 feet

GPI

PROJECT NO.: 3015.I

BELLA TERRA RESIDENTIAL

LOG OF BORING NO. B-2

FIGURE B-2

MOISTURE (%)	DRY DENSITY (PCF)	PENETRATION RESISTANCE (BLOWS/FOOT)	SAMPLE TYPE	DEPTH (FEET)	DESCRIPTION OF SUBSURFACE MATERIALS	ELEVATION (FEET)
				0	This summary applies only at the location of this boring and at the time of drilling. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The data presented is a simplification of actual conditions encountered.	
					4 inches AC over 4 inches AB	
16.1	108	9	D		Fill: **SILTY SAND (SM)** grey-brown, moist, loose, mottled, trace clay	20
34.4		1	D	5	Natural: **CLAY (CL)** grey, wet, soft	
56.1	66	1	D		@6 1/2 feet, sand lens	15
25.4	95	12	D	10	**SAND (SP)** light grey-brown, wet, medium dense, fine to medium grained	
					CLAY (CL) dark grey, wet	10
33.7	75	3	D	15	@ 15 feet, peat lens	
					SILTY SAND (SM) grey, wet, loose	5
34.8	78	3	D	20	**CLAY (CL)** dark grey to black, wet, soft, some organics	
					SILTY SAND (SM) grey, wet, loose	
					CLAY (CL) grey, wet, stiff, interbedded thin layers of silty sand	0
38.7	82	5	D	25		
						-5
34.3	87	8	D	30		
						-10
44.7	75	4	D	35		
						-15

DRAFT

SAMPLE TYPES
- C Rock Core
- S Standard Split Spoon
- D Drive Sample
- B Bulk Sample
- T Tube Sample

DATE DRILLED: 8-3-00

EQUIPMENT USED: 6" Rotary Wash

GROUNDWATER LEVEL:(ft): 6 1/2 feet

GPI

PROJECT NO.:3015.I

BELLA TERRA RESIDENTIAL

LOG OF BORING NO. B-3

FIGURE B-3

	MOISTURE (%)	DRY DENSITY (PCF)	PENETRATION RESISTANCE (BLOWS/FOOT)	SAMPLE TYPE	DEPTH (FEET)	DESCRIPTION OF SUBSURFACE MATERIALS	ELEVATION (FEET)
						This summary applies only at the location of this boring and at the time of drilling. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The data presented is a simplification of actual conditions encountered.	
	35.7	1	5	D	40		
						Total Depth 41 feet	-20
						Note: Boring also designated as B-3 in GPI's 2002 investigation for Bella Terra (Ref. 1)	

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SAMPLE TYPES
- C Rock Core
- S Standard Split Spoon
- D Drive Sample
- B Bulk Sample
- T Tube Sample

DATE DRILLED: 8-3-00

EQUIPMENT USED: 6" Rotary Wash

GROUNDWATER LEVEL:(ft): 6 1/2 feet

GPI

PROJECT NO.: 3015.I
BELLA TERRA RESIDENTIAL

LOG OF BORING NO. B-3

FIGURE B-3

MOISTURE (%)	DRY DENSITY (PCF)	PENETRATION RESISTANCE (BLOWS/FOOT)	SAMPLE TYPE	DEPTH (FEET)	DESCRIPTION OF SUBSURFACE MATERIALS	ELEVATION (FEET)
					This summary applies only at the location of this boring and at the time of drilling. Subsurface conditions may differ at other locations and may change at this location with the passage of time. The data presented is a simplification of actual conditions encountered.	
			B	0	5 inches AC over 4 inches AB	
					Fill?: Interbedded **SILTY SAND (SM)** and **CLAY (CL)** brown and light brown, moist, mottled	
26.9	89	4	D			20
					Natural: **SILTY SAND (SM)** light brown and grey, moist, loose, interbedded clay lenses	
38.1	75	2	D	5	**CLAY (CL)** dark grey, wet, firm, some sandy silt lenses	
26.8	95	9	D		**SAND (SP)** light brown, wet, loose, medium grained	15
64.5	50	2	D	10	@ 10 feet, lens of peat	
					SILTY CLAY (CL) grey, wet, soft, some organics	10
30.0	90	12	D	15	**SILTY SAND (SM)** grey, wet, medium dense	
					CLAY (CL) dark grey, wet, firm	5
26.0	95	11	D	20	**SILTY SAND (SM)** grey, wet, medium dense	
					CLAY (CL) grey, wet, stiff	0
42.1	78	5	D	25		-5
54.2	71	11	D	30		
					Total Depth 31 feet	

Note: Boring formerly designated as B-6 in GPI's 2002 investigation for Bella Terra (Ref. 1)

DRAFT

SAMPLE TYPES
- C Rock Core
- S Standard Split Spoon
- D Drive Sample
- B Bulk Sample
- T Tube Sample

DATE DRILLED: 8-3-00

EQUIPMENT USED: 6" Rotary Wash

GROUNDWATER LEVEL:(ft): 6 feet

GPI

PROJECT NO.: 3015.1
BELLA TERRA RESIDENTIAL

LOG OF BORING NO. B-4

FIGURE B-4

APPENDIX C

APPENDIX C

LABORATORY TESTS

INTRODUCTION

Representative undisturbed soil samples and bulk samples were carefully packaged in the field and sealed to prevent moisture loss. The samples were then transported to our Cypress office for examination and testing assignments. Laboratory tests were performed on selected representative samples as an aid in classifying the soils and to evaluate the physical properties of the soils affecting foundation design and construction procedures. Detailed descriptions of the laboratory tests are presented below under the appropriate test headings. Test results are presented in the figures that follow.

MOISTURE CONTENT AND DRY DENSITY

Moisture content and dry density were determined from a number of the ring samples. The samples were first trimmed to obtain volume and wet weight and then were dried in accordance with ASTM D 2216. After drying, the weight of each sample was measured, and moisture content and dry density were calculated. Moisture content and dry density values are presented on the boring logs in Appendix B.

PERCENT PASSING NO. 200 SIEVE

A total of five soil samples were dried, weighed, soaked in water until individual soil particles were separated, and then washed on the No. 200 sieve. That portion of the material retained on the No. 200 sieve was oven-dried and weighed to determine the percentage of the material passing the No. 200 sieve. The percentages passing the No. 200 sieve are tabulated below.

BORING NO.	DEPTH (ft)	SOIL DESCRIPTION	PERCENT PASSING No. 200 SIEVE
B-1	50	Clayey Silt (ML)	90
B-2	0-1	Silty Sand (SM)	29
B-3	10	Sand (SP)	2
B-3	30	Sandy Silt (ML)	82
B-4	20	Silty Sand (SM)	37

DIRECT SHEAR

Direct shear tests were performed on undisturbed samples in accordance with ASTM D 3080. The samples were placed in the shear machine, and a normal load comparable to the in-situ overburden stress was applied. The samples were inundated, allowed to consolidate, and then were sheared to failure. The tests were repeated on additional test specimens under increased normal loads. Shear stress and sample deformation were monitored throughout the test. The results of the direct shear tests are presented in Figure C-1.

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CONSOLIDATION

A one-dimensional consolidation test was performed on undisturbed samples in accordance with ASTM D 2435. After trimming the ends, the sample was placed in the consolidometer and loaded to up to 0.4 ksf. Thereafter, the sample was incrementally loaded to a maximum load of up to 25.6 ksf. The sample was inundated at 1.6 ksf. Sample deformation was measured to 0.0001 inch. Rebound behavior was investigated by unloading the sample back to 0.4 ksf. Results of the consolidation tests, in the form of percent consolidation versus log pressure are presented in Figures C-2 and C-3.

COMPACTION TEST

Maximum dry density/optimum moisture tests were performed in accordance with ASTM D 1557 on a representative bulk sample of the surficial soils. The test result is as follows:

BORING/ TEST PIT NO.	DEPTH (ft)	SOIL DESCRIPTION	MAXIMUM DRY DENSITY (pcf)	OPTIMUM MOISTURE (%)
B-2	0-1	Silty Sand (SM)	119	12.0

R-VALUE

Suitability of the near-surface soils for pavement was evaluated by conducting an R-value test. The test was performed in accordance with ASTM D2844. The results of the test are as follows:

BORING NO.	DEPTH (ft)	SOIL DESCRIPTION	R-VALUE
B-4	0-1	Clayey Silt with Sand (ML)	30

EXPANSION INDEX

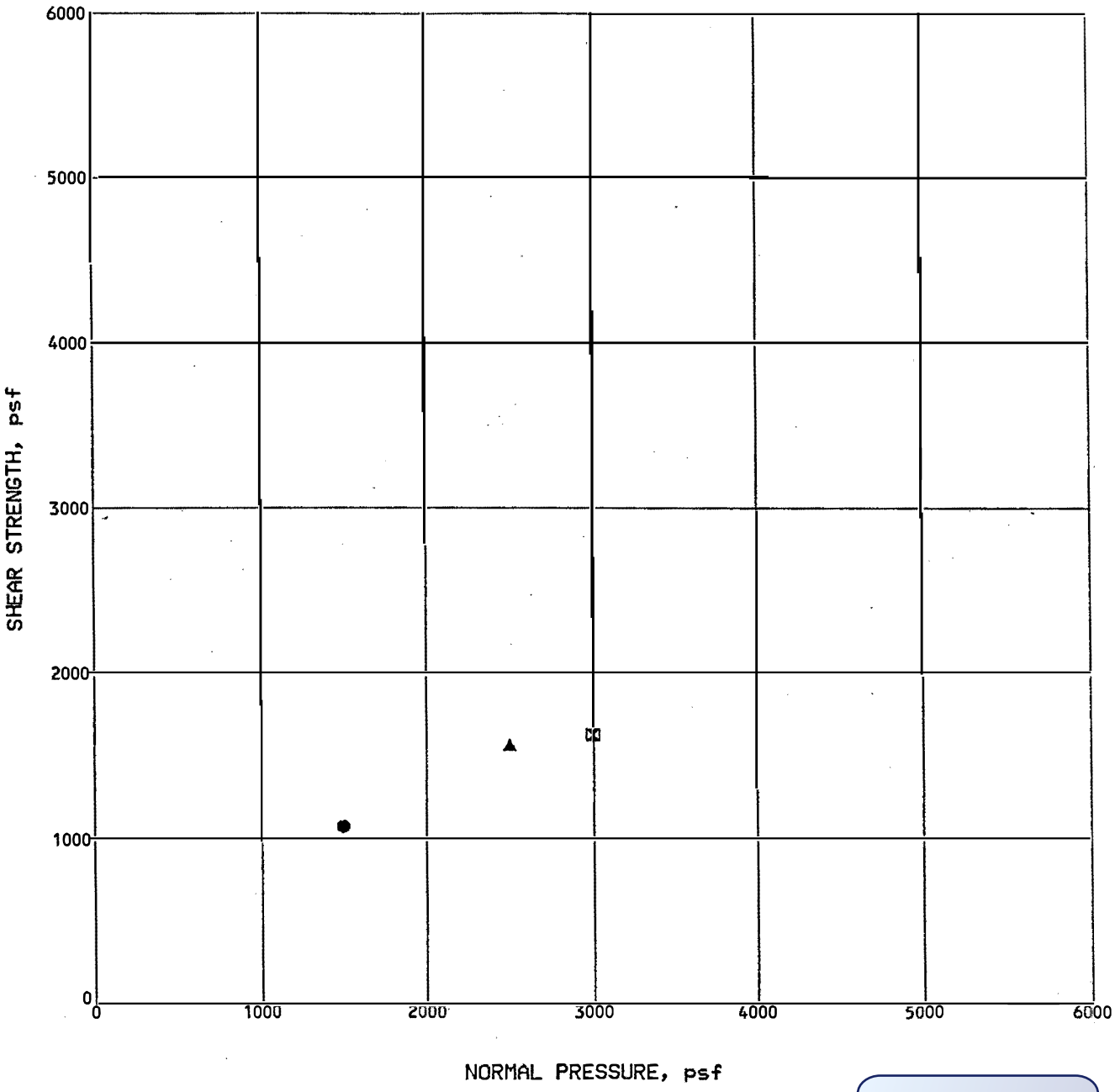
Expansion index tests were performed on bulk samples. The test was performed in accordance with ASTM 4289 to assess the expansion potential of on-site soils. The results of the test are summarized below:

BORING/ TEST PIT NO.	DEPTH (ft)	SOIL DESCRIPTION	EXPANSION INDEX
B-1	2	Silty Clay/Sandy Clay (CL)	74

CORROSIVITY

Soil corrosivity testing was performed by M.J. Schiff and Associates on a soil sample provided by GPI. The test results are summarized in Table 1 of this appendix.

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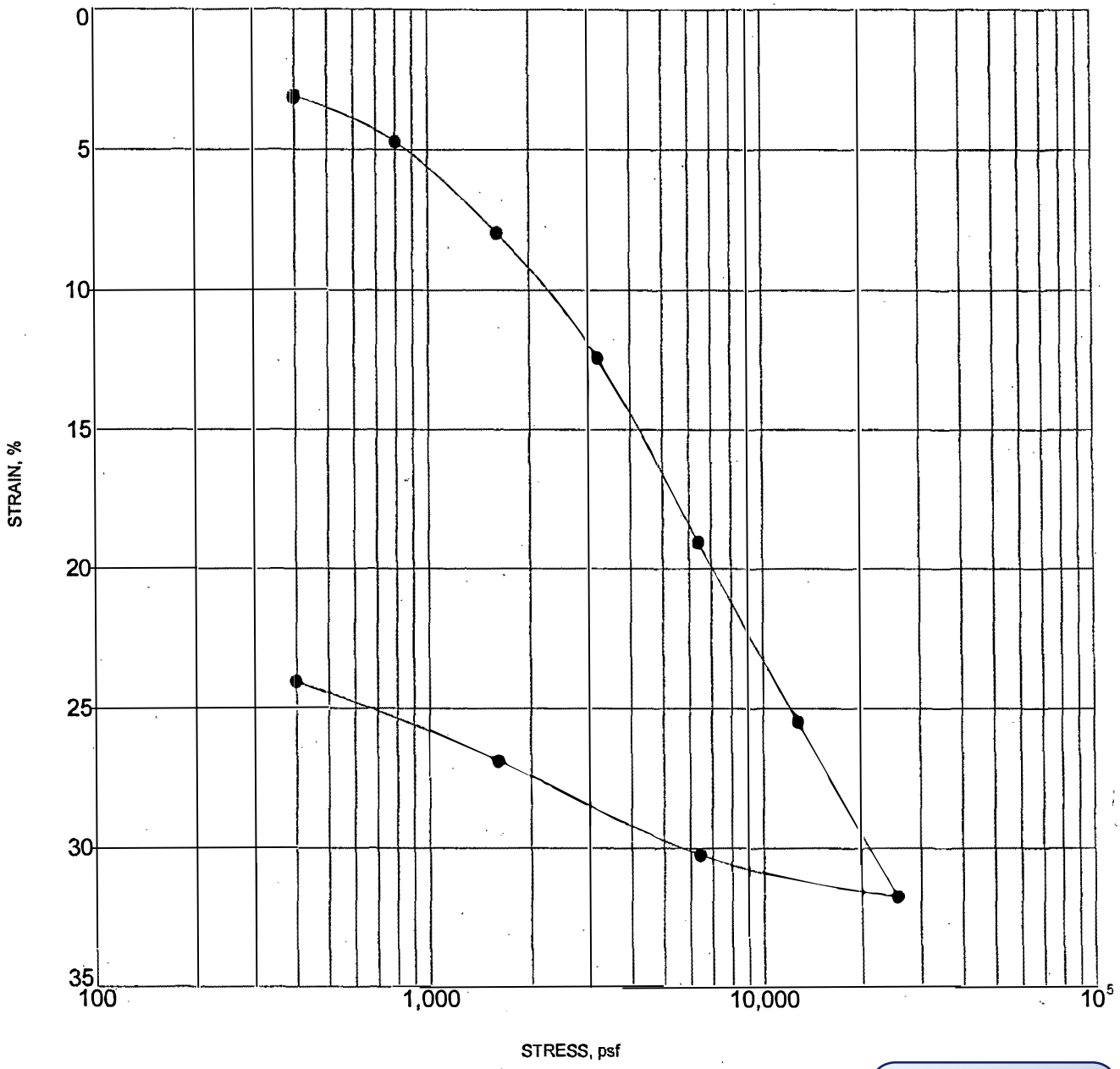
	Sample Location		Classification	DD,pcf	MC,%
●	B-3	20.0	CLAY (CL)	72	35
⊠	B-3	35.0	CLAY (CL)	76	45
▲	B-4	30.0	SILTY CLAY (CL)	65	54

PROJECT: BELLA TERRA RESIDENTIAL PROJECT NO.: 3015.I

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DIRECT SHEAR TEST RESULTS

FIGURE C-1



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Sample inundated at 400 psf

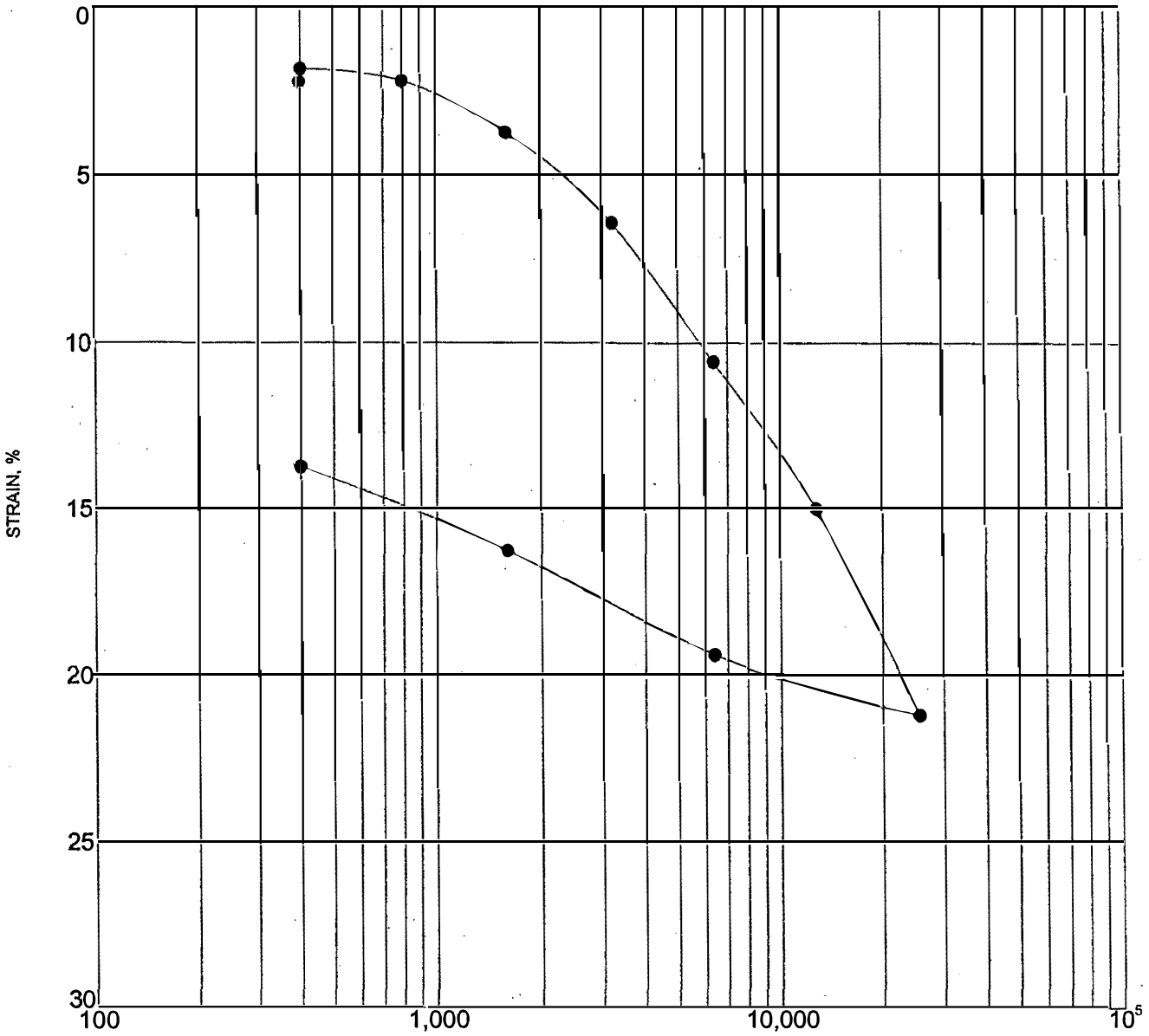
Sample Location		Classification	DD,pcf	MC,%
● B-2	10.0	SILTY CLAY (CL)	69	49.9

PROJECT: BELLA TERRA RESIDENTIAL PROJECT NO.: 3015.I

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CONSOLIDATION TEST

FIGURE C-2



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Sample inundated at 400 psf

	Sample Location		Classification	DD,pcf	MC,%
●	B-1	25.0	SILTY CLAY (CL)	79	40.6

PROJECT: BELLA TERRA RESIDENTIAL PROJECT NO.: 3015.I

GPI

CONSOLIDATION TEST

FIGURE C-3

M. J. Schiff & Associates, Inc.
Consulting Corrosion Engineers - Since 1959
1308 Monte Vista Avenue, Suite 6
Upland, CA 91786-8224
Phone: 909/931-1360

Table 1 - Laboratory Tests on Soil Samples

MJS&A #00-0501LAB
15-Aug-00

Sample ID			B-4 @ 2' Silty Sand (SM)	B-2/B-3 Mix @ 10-15' Silty Clay (CL)
Resistivity		**Units**		
as-received		ohm-cm	1,200	470
saturated		ohm-cm	970	470
pH			7.0	6.4
Electrical				
Conductivity		mS/cm	0.29	3.70
Chemical Analyses				
Cations				
calcium	Ca^{2+}	mg/kg	180	5,122
magnesium	Mg^{2+}	mg/kg	19	306
sodium	Na^{1+}	mg/kg	80	ND
Anions				
carbonate	CO_3^{2-}	mg/kg	ND	ND
bicarbonate	HCO_3^{1-}	mg/kg	305	46
chloride	Cl^{1-}	mg/kg	71	440
sulfate	SO_4^{2-}	mg/kg	340	12,300
Other Tests				
ammonium	NH_4^{1+}	mg/kg	na	na
nitrate	NO_3^{1-}	mg/kg	na	na
sulfide	S^{2-}	qual	na	na
Redox		mv	na	na

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Electrical conductivity in millisiemens/cm and chemical analysis were made on a 1:5 soil-to-water extract.
mg/kg = milligrams per kilogram (parts per million) of dry soil.
Redox = oxidation-reduction potential in millivolts
ND = not detected
na = not analyzed

APPENDIX D

APPENDIX D

SITE-SPECIFIC RESPONSE SPECTRA

Site specific response spectra were generated in accordance with the 2019 California Building Code (CBC) (Section 1613A) and Section 21.2 of ASCE 7-16 (ASCE, 2017), as well as ASCE 7-16 Supplement 1 (2018). Creation of a site-specific response spectrum requires analyzing site-specific deterministic and probabilistic seismic response spectra in order to create the Risk-Targeted Maximum Considered Earthquake (MCE_R) and Design response spectra.

We calculated the deterministic and probabilistic site-response spectra using web-based tools that estimate uniform hazard spectra using faults as earthquake sources. The web tools include geographic and seismic information on known active faults in California based on the 2014 USGS fault model. For both our deterministic and probabilistic analyses, we used four 2014 NGA West 2 attenuation relationships to determine the geometric-mean horizontal component of ground motion: Abrahamson-Silva-Kamai (2014), Boore-Stewart-Seyhan-Atkinson (2014), Campbell-Bozorgnia (2014), and Chiou-Youngs (2014).

For our evaluations, we used a shear wave velocity, V_{S30}, of approximately 248 meters per second (m/s), or about 813 feet per second (fps), for the selected attenuation relationships. This value corresponds to the lower range of the CBC Site Class D (stiff soil) and was directly measured in our CPT C-4 to refusal at a depth of 80 feet during our investigation at the project site.

Probabilistic Spectra

The probabilistic (MCE_R) ground motion spectra (per the Method 1 requirements of Section 21.2.1.1, ASCE 7-16) were calculated using the USGS Unified Hazard Tool website. Using inputs of the site coordinates, Site Class, and time horizon (return period), the web tool outputs the Uniform Hazard Response Spectrum (UHRS) for predetermined Site Classes and shear wave velocities. For our analysis, we utilized the *Dynamic: Conterminous U.S. 2014 (update, v4.2.0)* edition of the web tool. We calculated response spectra for 259 m/s (Site Class D) and 180 m/s (D/E Boundary) and then linearly interpolated between those spectra based on our estimated site-specific shear wave velocity of about 248 m/s.

The MCE_R corresponds to an earthquake ground motion having a 2 percent probability of exceedance within a 50-year period, or an average return period of 2,475 years. The final probabilistic response spectrum was based on the geometric mean horizontal component, scaled by factors to convert the geometric-mean response to the maximum-rotated response, of the spectral response values at 5% damping for the four above noted attenuation relationships. The maximum rotated component (MRC) response factors used were based on the period dependent factors developed by Huang, Whittaker, and Luco (2008) and presented in Section 21.2 of ASCE 7-16. The weighted average, maximum-rotated site-specific probabilistic response for the above predetermined shear wave velocities, as well as the final interpolation based on the site-specific shear wave velocity, are shown on Figure D-1 (see attached).

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Deterministic Spectra

Site-specific deterministic MCE response spectra were generated per the requirements of ASCE Section 7-16. The response spectrum was generated for nearby active faults, which were determined based on a combination of proximity, previous screening using the EZ-Frisk program, and the table of deaggregation contributors developed with the USGS Unified Hazard Tool. Based on the above resources, the controlling deterministic response spectrum is predominantly based on Newport-Inglewood Fault (Alt 1).

Spectral acceleration ordinates were calculated utilizing the Pacific Earthquake Engineering Research Center (PEER) ground motion database and the PEER NGA-West2 Spectrum model. We utilized the four previously noted attenuation relationships (equally weighted) and determined the required input fault parameters from USGS web resources (see references). Per the requirements of ASCE 7-16, we utilized an epsilon value of 1.0 for our analysis, which corresponds to the 84th percentile of the geometric-mean component (S_a + one standard deviation) of the spectral acceleration at 5% damping. As with the probabilistic spectrum, the geometric-mean values were scaled by period-dependent factors per Huang et al (2008) to obtain the maximum-rotated response. The site-specific deterministic response spectrum is shown on Figure D-2.

MCE_R and Design Response Spectra

The above described analytical steps are presented in the attached Table D-1, Risk-Targeted Site-Specific Seismic Response Spectra Worksheet.

The site-specific MCE_R response spectrum was generated per the requirements of Section 21.2 of ASCE 7-16 by comparing the spectral response accelerations from the probabilistic MCE_R (Section 21.2.1, see Figure 1) and the deterministic MCE_R (Section 21.2.2, see Figure 2), with the resulting MCE_R response spectrum being the lesser of the spectra accelerations at each period. The ordinates for the MCE_R response spectrum are presented in Table D-1 (Column 11).

The site-specific design response spectrum was generated per the requirements by taking 2/3 of the risk-targeted MCE_R response spectrum, but confirming that the values are not less than 80 percent of the spectral acceleration determined per Sections 11.4.6 and 21.3 of ASCE 7-16. The ordinates for the site-specific design response spectrum are presented in Table D-1 (Column 12).

The risk-targeted site-specific MCE_R and design response spectra, as well as the mapped CBC response spectrum, are shown on Figure D-3 and tabulated in Table D-1.

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REFERENCES

American Society of Civil Engineers (2017), "Minimum Design Loads and Associated Criteria for Buildings and Other Structures," ASCE/SEI 7-16.

American Society of Civil Engineers (2018), "Minimum Design Loads and Associated Criteria for Buildings and Other Structures, Supplement 1" ASCE/SEI 7-16, effective December 12, 2018.

California Office of Statewide Health Planning and Development (OSHPD), Seismic Design Maps Website, https://seismicmaps.org/

Huang, Y. N., Whittaker, A. S., and Luco, N. (2008). "Maximum spectral demands in the near-fault region." *Earthquake Spectra,* 24(1), 319-341.

Pacific Earthquake Engineering Research Center (PEER) Ground Motion Database, NGA-West2 Shallow Crustal Earthquakes in Active Tectonic Regimes, Target Spectrum (used for deterministic site-specific seismic analysis), accessed December 2020, https://ngawest2.berkeley.edu/spectras/new?sourceDb_flag=1

United States Geological Survey (USGS), 2008 National Seismic Hazard Maps, Source Parameters, accessed December 2020, http://geohazards.usgs.gov/

United States Geologic Survey (USGS), M 7.2 Scenario Earthquake – Newport-Inglewood alt 1, accessed December 2020, https://earthquake.usgs.gov/scenarios/eventpage/bssc2014newportinglewoodalt1_m7p15_se/executive

United States Geologic Survey (USGS), M 7.2 Scenario Earthquake – Newport-Inglewood alt 2, accessed December 2020, https://earthquake.usgs.gov/scenarios/eventpage/bssc2014newportinglewoodalt2_m7p15_se/executive

United States Geologic Survey (USGS), Unified Hazard Tool website, accessed December 2020, https://earthquake.usgs.gov/hazards/interactive/

United States Geologic Survey (USGS), USGS Earthquake Scenario Map website (Building Seismic Safety Council, BSSC 2014), accessed December 2020, https://usgs.maps.arcgis.com/apps/webappviewer/index.html?id=14d2f75c7c4f4619936dac0d14e1e468

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TABLE D-1
RISK TARGETED SITE-SPECIFIC SEISMIC REPONSE WORKSHEET

Project	**BELLA TERRA**
Proj. No.	**3015.I**
Latitude	**33.7322**
Longitude	**-117.9949**

Site Class	**D**
T_o	0.129 sec
T_s	0.646 sec
T_L	8.0 sec
V_S	813 ft/sec

2019 CBC Parameters

S_s	1.402	S_1	0.504
F_a	1.000	F_v*	1.796
S_{MS}	1.402	S_{M1}	0.905
S_{DS}	0.935	S_{D1}	0.603
0.08 F_v/F_a	0.144	0.4F_v/F_a	0.718

PGA_M	0.665
C_{RS}	0.912
C_{R1}	0.917

NGA West2 Attentuation Relationships

1) Abrahamson-et al (2014)
2) Boore-et al (2014)
3) Cambell-Bozorgnia (2014)
4) Chiou-Youngs (2014)

SITE-SPECIFIC PARAMETERS

S_{MS}	**1.685**
S_{M1}	**1.472**
S_{DS}	**1.124**
S_{D1}	**0.981**
PGA_M	**0.732**

1)	2)	3)	4)	5)	6)	7)	8)	9)	10)	11)	12)
Period (sec)	2019 CBC MCE_R Spectrum	2019 CBC Design Response Spectrum	Risk Coefficient C_R	Scaled MCE_R Deterministic Spectrum (if required)	Probabilistic MCE_R Spectrum	Probabilistic w/ Risk Coefficient C_R	84th Percentile Deterministic Spectrum	2/3 Site Specific MCE_R Spectrum	80% of 2019 CBC Design Spectrum	**Site Specific MCE_R Spectrum**	**Design Response Spectrum**
0.000	0.665	0.443	0.912	NA	0.802	0.732	0.818	0.488	0.355	**0.732**	**0.488**
0.050	0.887	0.591	0.912	NA	1.080	0.985	0.887	0.592	0.424	**0.887**	**0.592**
0.100	1.212	0.808	0.912	NA	1.357	1.238	1.211	0.807	0.549	**1.211**	**0.807**
0.129	1.402	0.935	0.912	NA	1.491	1.360	1.342	0.895	0.621	**1.342**	**0.895**
0.200	1.402	0.935	0.912	NA	1.817	1.657	1.662	1.105	0.748	**1.657**	**1.105**
0.300	1.402	0.935	0.913	NA	2.052	1.873	1.991	1.248	0.748	**1.873**	**1.248**
0.400	1.402	0.935	0.913	NA	2.032	1.855	2.115	1.237	0.748	**1.855**	**1.237**
0.500	1.402	0.935	0.914	NA	2.010	1.837	2.116	1.225	0.748	**1.837**	**1.225**
0.600	1.402	0.935	0.915	NA	1.875	1.715	2.012	1.143	0.748	**1.715**	**1.143**
0.646	1.402	0.935	0.915	NA	1.813	1.659	1.965	1.106	0.748	**1.659**	**1.106**
0.700	1.293	0.862	0.915	NA	1.740	1.592	1.908	1.061	0.748	**1.592**	**1.061**
0.750	1.207	0.805	0.915	NA	1.672	1.531	1.856	1.020	0.748	**1.531**	**1.020**
0.800	1.131	0.754	0.916	NA	1.621	1.485	1.815	0.990	0.748	**1.485**	**0.990**
0.900	1.006	0.671	0.916	NA	1.520	1.393	1.735	0.928	0.747	**1.393**	**0.928**
1.000	0.905	0.603	0.917	NA	1.418	1.300	1.654	0.867	0.672	**1.300**	**0.867**
2.000	0.453	0.302	0.917	NA	0.803	0.736	0.973	0.491	0.336	**0.736**	**0.491**
3.000	0.302	0.201	0.917	NA	0.532	0.488	0.659	0.326	0.224	**0.488**	**0.326**
4.000	0.226	0.151	0.917	NA	0.382	0.350	0.460	0.233	0.168	**0.350**	**0.233**
5.000	0.181	0.121	0.917	NA	0.295	0.271	0.343	0.180	0.134	**0.271**	**0.180**

(Based Upon Chapters 11 and 21 of ASCE 7-16; 04-14-2020 DJB)

TABLE D-1
RISK TARGETED SITE-SPECIFIC SEISMIC REPONSE WORKSHEET

EXPLANATION: NOTES AND REFERENCES

INPUT BLUE ONLY - RED AND BLACK CALCULATED

Column Descriptions

01) Periods including To and Ts calculated from Section 11.4.6 (ASCE 7-16)

02) OSHPD, Seismic Design Maps Web Application - MCE_R Response Spectrum (seismicmaps.org) and Section 11.4.7 (7-16)

03) OSHPD, Seismic Design Maps Web Application - Design Spectrum (2/3 of Column 2) per Section 11.4.6 (7-16)

04) Risk Coefficient, C_R, for 0.2s and 1.0s periods (ASCE 7-16, Section 21.2.1.1); from OSHPD web application

05) Deterministic Lower Limit on MCE_R if required (ASCE 7-16 Supplement 1; Section 21.2.2)

06) USGS Unified Hazard Tool (UHT), 2% in 50 years Probabilistic Spectrum; scaled w/ MRC factors per Huang et al (2008); per ASCE 7-16, Section 21.2.1.1

07) USGS UHT, Probabilistic MCE_R Spectrum: Product of 2% in 50yr Spectrum and Risk Coefficient (Col. 4 * Col. 6); (ASCE 7-16, Section 21.2.1.1)

08) PEER Ground Motion Database, 84th Percentile Deterministic Spectrum; controlling fault source (ASCE 7-16; Section 21.2.2)

09) Uncorrected Design Response Spectrum (ASCE 7-16 Sec. 21.3), 2/3 * Lesser of Col. 7 & Greater of Cols. 5 & 8 (not less than 80% PGAM per Sec. 21.5.3)

10) 80% of 2019 CBC Design Spectra (Column 3), (ASCE 7-16, Section 21.3) Lower Limit of the Design Spectrum

11) Site-Specific MCE_R (ASCE 7-16, Section 21.2.3); 150% of Design Response Spectrum (Column 12)

12) Final Design Response Spectrum (ASCE 7-16, Section 21.3); Greater of Columns 9 and 10

T_L = Figure 22-12 ASCE 7-16 (typically 8 sec Southern California)

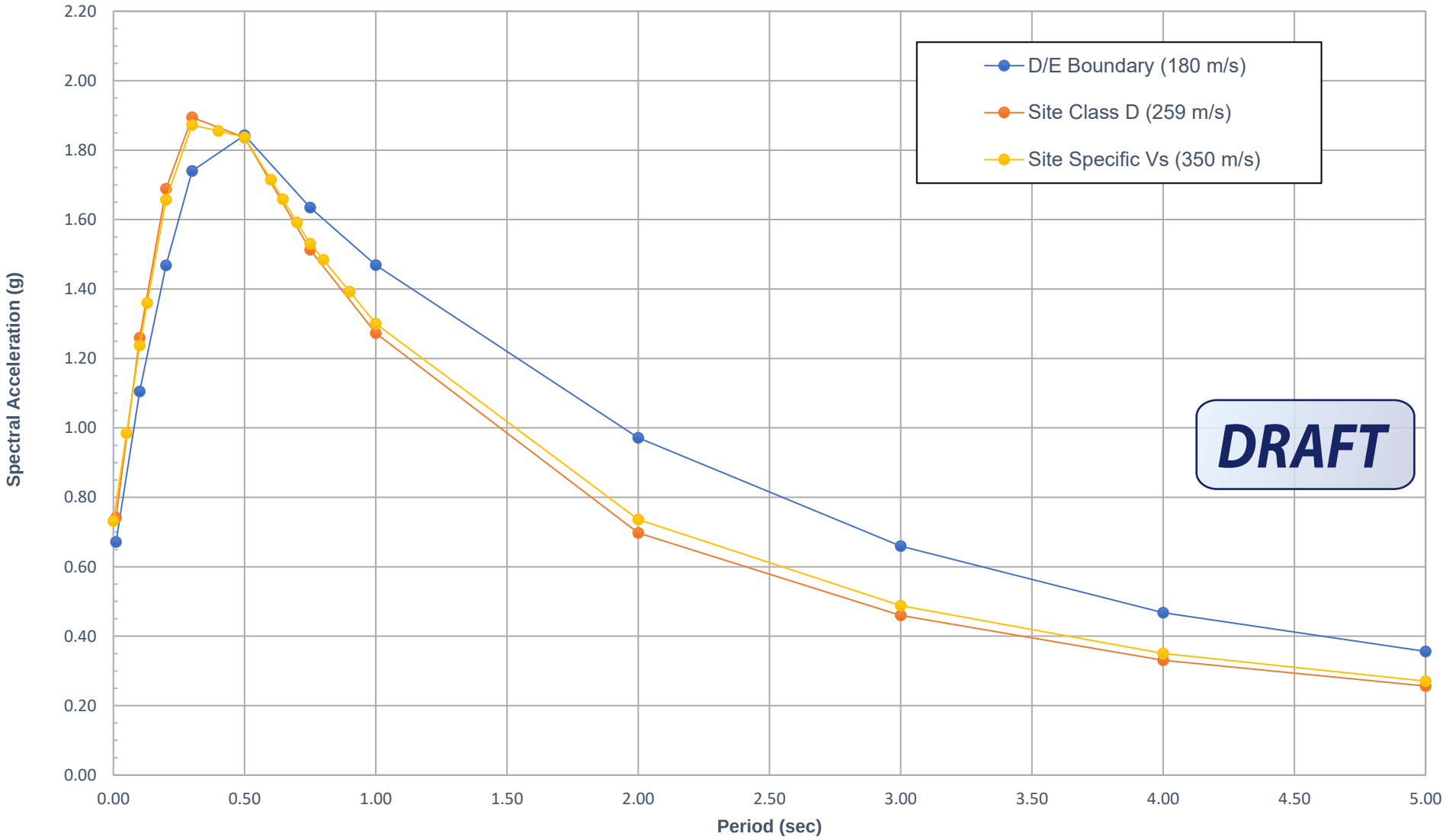
MUST CHECK THAT VALUES EXCEED MINIMUMS

Minimum Allowable Value of MCE PGA (Column 9):	**0.532**
(80% of PGA_M)	
Value of S_{DS}:	**1.124**
(Maximum of 90% of Design S_a at any period)	
Value of S_{D1}:	**0.981**
(Maximum of $T*S_a$ for periods from 1 to 5 seconds)	

* = F_V is modified for the deterministic lower limit determinations (Fig. 21.2-1) based on the requirements of Section 11.4.8 and the Site Specific Ground Motion Hazard Analyses as detailed in Section 21.3

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(Based Upon Chapters 11 and 21 of ASCE 7-16; 04-14-2020 DJB)

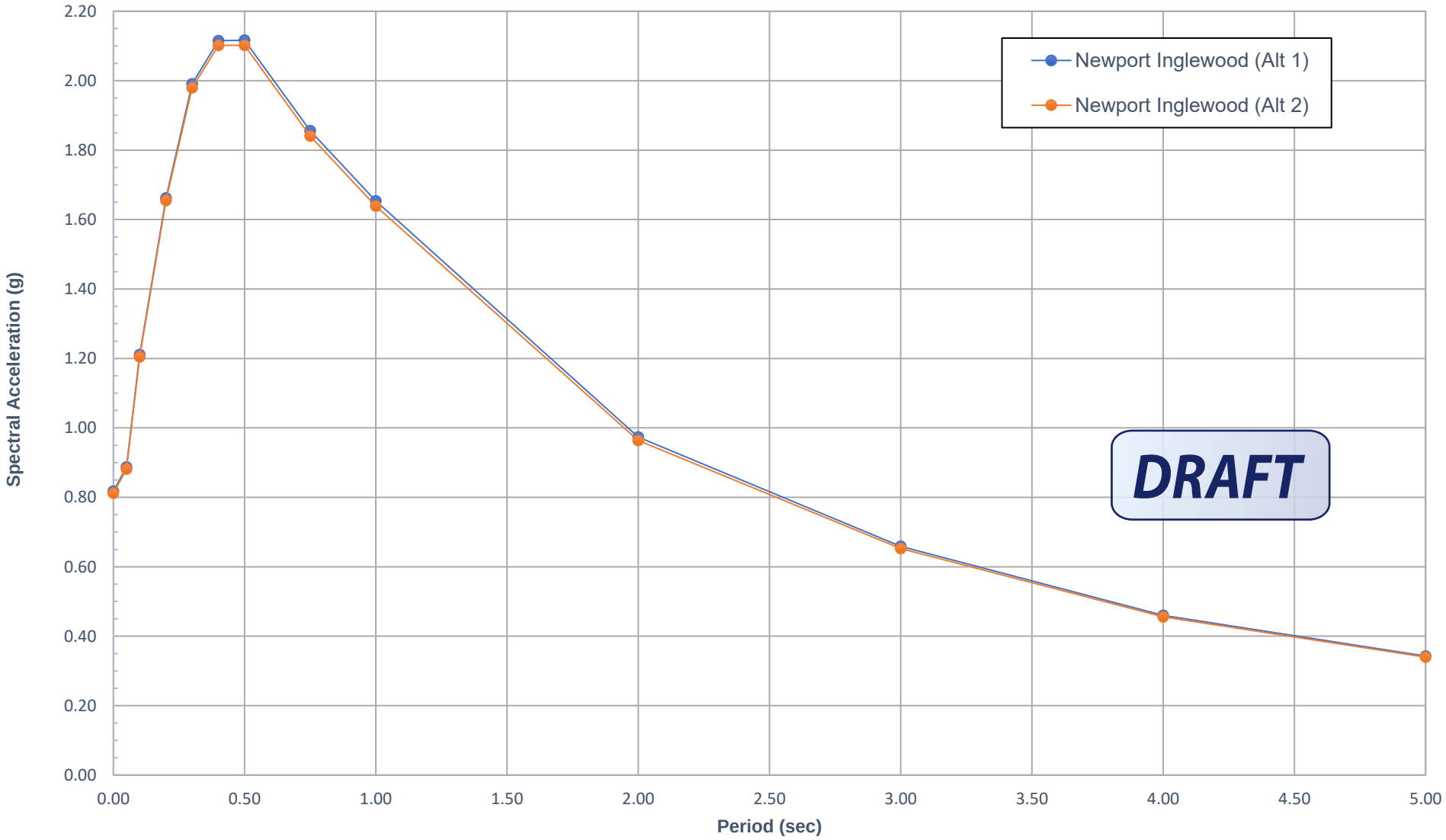


Site-Specific Probabilisitic MCE_R Response Spectra @ 5% Damping
per Chapter 21; ASCE 7-16 (2019 CBC)
Bella Terra
Huntington Beach, California

GEOTECHNICAL PROFESSIONALS, INC.

Project No. 3015.I

Figure D-1

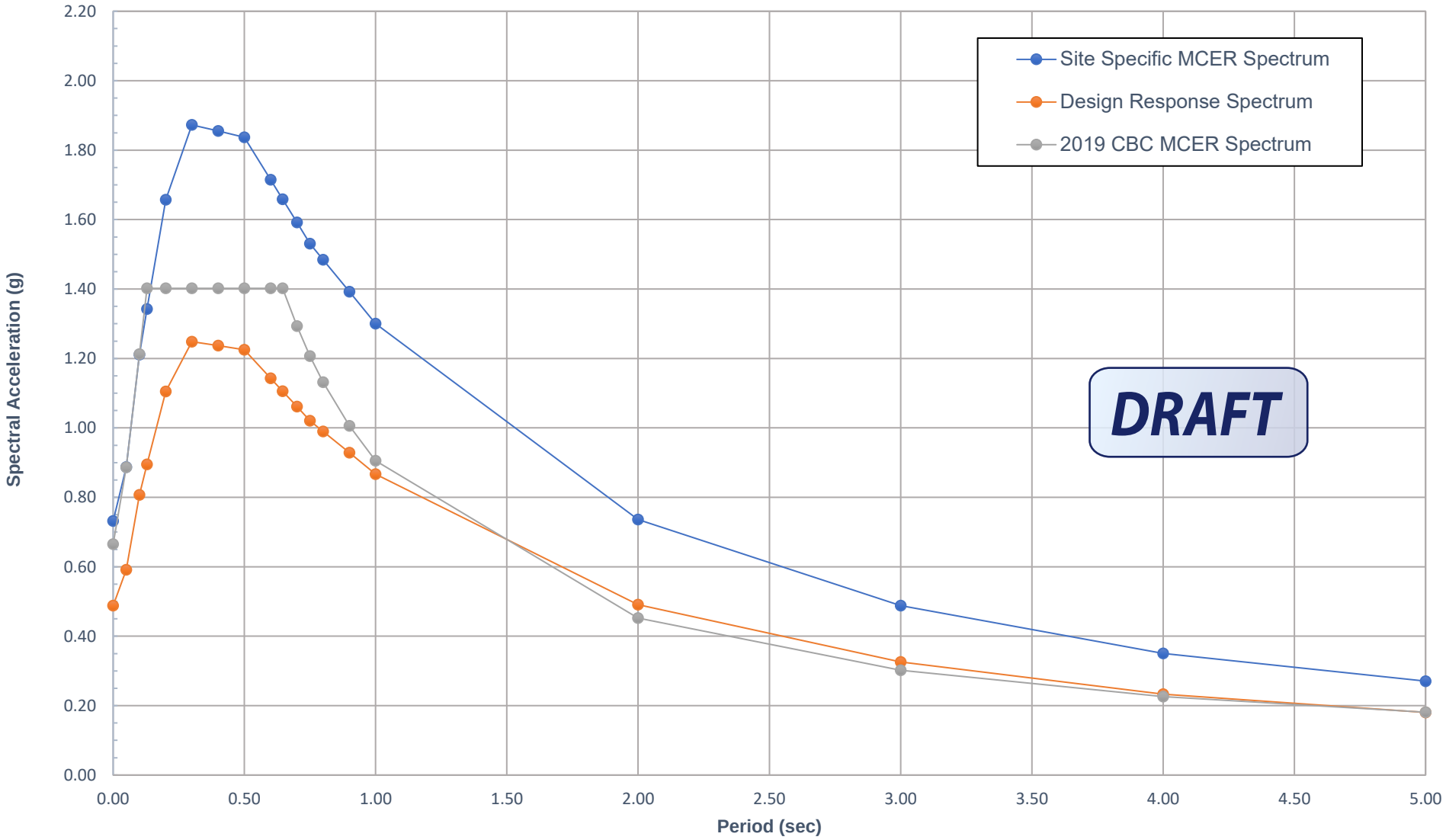


Site-Specific 84th Percentile Deterministic MCE_R Response Spectra
per Chapter 21; ASCE 7-16 (2019 CBC)
Bella Terra
Huntington Beach, California

GEOTECHNICAL PROFESSIONALS, INC.

Project No. 3015.I

Figure D-2



Site-Specific MCE_R and Design Response Spectra @ 5% Damping
per Chapter 21; ASCE 7-16 (2019 CBC)
Bella Terra
Huntington Beach, California

GPI GEOTECHNICAL PROFESSIONALS, INC.

Project No. 3015.I

Figure D-3